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Frontiers of molten metal technology: Fundamentals, mechanics, and emerging applications

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Highlights

- Molten metals merge fluidity with metallic bonding across diverse applications.
- Electronic structure, thermophysics, and oxidation govern metal interfaces.
- Wetting, capillarity, and Marangoni effects control interfacial behavior.
- Self-healing, electrocapillarity, and dynamic transitions define unusual interfaces.
- Confinement and soft-matter interactions enable reconfigurable molten-metal systems.

Abstract

Molten metals retain metallic bonding in the liquid state, giving rise to interfacial properties that differ fundamentally from those of conventional solids and molecular liquids. Understanding these unique interfaces is increasingly important as molten-metal systems transition from traditional metallurgical processes to advanced functional technologies. This study provides a comprehensive review of the interfacial science governing molten metals and their evolution toward emerging applications. The discussion begins with the fundamental physicochemical characteristics of molten metals, emphasizing the role of liquid-state electronic structure, surface oxidation, and thermophysical properties in controlling interfacial stability and reactivity. The thermodynamics of molten-metal interfaces are further discussed, including wetting behavior, interfacial free energy, capillary effects, and Marangoni-driven flows that regulate dynamic interactions at liquid–solid, liquid–liquid, and liquid–gas boundaries. Particular attention is given to unconventional interfacial phenomena such as reversible liquid–solid transitions, self-healing behavior, electrocapillarity, and shape reconfiguration, which extend beyond classical interface theories. The stabilizing and functional roles of surface oxide layers are also analyzed, highlighting how oxide skins govern mechanical integrity, chemical activity, and surface adaptability. Furthermore, molten-metal behavior under confinement and in contact with soft materials is discussed, revealing size-dependent effects and mechanically reconfigurable interfaces. Finally, the implications of these interfacial mechanisms are evaluated across emerging technological domains, including soft electronics, energy conversion and storage, catalysis, additive manufacturing, and thermal management. By integrating insights from metallurgy, surface science, and materials engineering, this critical review establishes a unified framework for understanding molten-metal interfaces and outlines future directions for their predictive control and scalable technological implementation.

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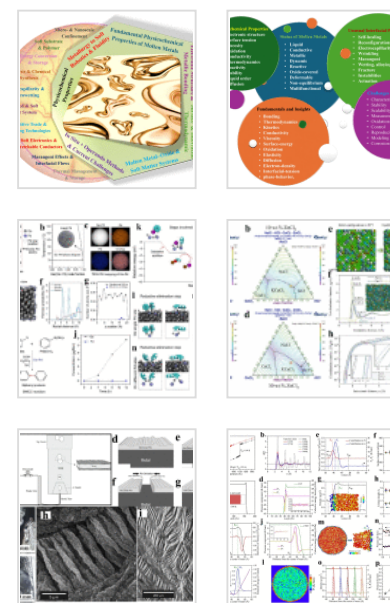
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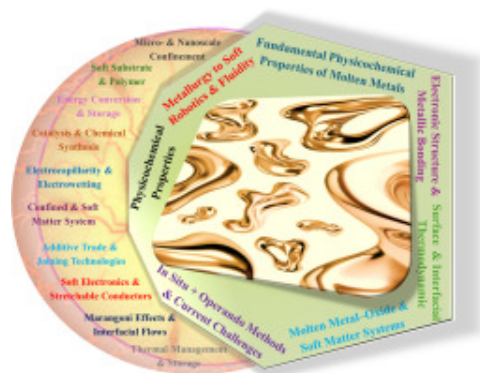
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Keywords

Molten metals; Liquid metal interfaces; Interfacial phenomena; Oxide skins; Wetting and capillarity; Electrocapillarity; Marangoni flow

Abbreviation

AFSD, Additive friction stir deposition; AIMD, Ab initio molecular dynamics; ALMPs, Adhesive liquid metal particles; AM, Additive manufacturing; BSE, Backscattered electrons; CERs, Cation-exchange resins; CF/MWCNTN, Carbon fiber/multi-walled carbon nanotube network; CMT, Cold metal transfer; CN, Coordination number; DOS, Density of states (eV^{-1}); EDX, Energy-dispersive X-ray spectroscopy; EGaIn–InSn, Eutectic gallium–indium–tin; EMI, Electromagnetic interference (dB); Ga–Pd, Gallium–Palladium; GC–MS, Gas chromatography–mass spectrometry; GO, Graphene oxide; ΔH_m , Enthalpy of melting; ΔH_v , Enthalpy of vaporization; ICP–MS, Inductively coupled plasma mass spectrometry; L, Length (mm); LM, Liquid metal; LMCT, Ligand-to-metal charge transfer; LMMPs, Liquid metal microparticles; MSO, Molten salt oxidation; M-4PyCN, Cyano-group-coordinated metal–organic interlayers; NMR, Nuclear magnetic resonance spectroscopy; OMO, Oxide–metal–oxide; PCCs, Phase change composites; PDMS,

Polydimethylsiloxane; PDF, Pair distribution function; ps, Picosecond; PV, Photovoltaic; PVA, Poly(vinyl alcohol); PW, Paraffin wax; SEI, Solid electrolyte interphase; SEM, Scanning electron microscopy; sEMG, Surface electromyography (mV or μV); SMCC, Suzuki–Miyaura C–C coupling; SMSI, Strong metal–support interaction; T_d , Droplet temperature (K); TEM, Transmission electron microscopy; TIG, Tungsten inert gas welding; T_s , Substrate temperature (K); TTM–MD, Two-temperature model molecular dynamics; XPS, X-ray photoelectron spectroscopy; XRD, X-ray diffraction (2θ , $^\circ$); 304SS, Cu/304 stainless steel; Physical Quantity, Length, mm, nm, μm , $\text{\AA}$; Temperature, $^\circ\text{C}$, K; Cohesive energy density, GPa; Density, g/cm^3 ; Energy / fluence, eV, J cm^{-2} ; Time, ns, ms, s; Concentration, at%, wt%, ppm; Electrical, mA, mV, μV ; Frequency / wavenumber, Hz, cm^{-1} ; Rotation speed, rpm; Velocity, mm/min; Angle (XRD), 2θ ($^\circ$); Electromagnetic interference, dB; Surface tension, mN/m; Tolman length, ($\text{\AA}$); Density for center of mass, $\text{\AA}$; Potential energy, eV/atom; Enthalpy of melting, (kJ/mol); Enthalpy of vaporization, kJ/mol; Note, All physical quantities are expressed in SI units.

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1. Introduction

Molten metals have played a central role in the development of human technology, traditionally serving as transient processing media rather than functional materials [1]. Since the Bronze Age, liquid metals have been valued for their ability to flow into molds and solidify into durable structures [2]. Consequently, the liquid state has traditionally been viewed merely as an intermediate phase whose significance ends once solidification occurs [3]. As a result, research has traditionally concentrated on the solid-state properties of metals, with comparatively little attention devoted to their liquid and interfacial behavior. This paradigm is now changing as advances in materials science increasingly demand materials that are adaptive, reconfigurable, and responsive, with

capabilities inherently linked to the liquid state [4], [5], [6], [7]. Molten metals signify a distinct class of materials that combine fluid dynamics with metallic bonding, high electrical conductivity, and extreme surface tension. Unlike molecular liquids, their cohesion arises from delocalized electrons, and unlike solids, they lack long-range crystalline order [8], [9], [10], [11], [12]. This duality gives rise to interfacial behaviors that deviate markedly from classical models of wetting, capillarity, and electrochemistry [13], [14]. In particular, molten metal interfaces are characterized by strong coupling between mechanical forces, charge redistribution, chemical reactions, and mass transport [15]. These synergisms give rise to interfacial phenomena that cannot be fully captured using equilibrium thermodynamics or static surface energy descriptions, necessitating a more dynamic and interdisciplinary framework [16], [17], [18].

Historically, mercury served as the primary model system for liquid metal research due to its room-temperature liquidity and electrical conductivity [19], [20]. Early studies leveraged mercury for electrochemical measurements, switches, and scientific instrumentation [21], [22], [23], [24]. However, its high toxicity and vapor pressure severely constrained its practical and biomedical use [25], [26], [27]. As a result, molten metals remained largely confined to high-temperature applications such as casting, soldering, and nuclear reactor cooling, where interfacial effects were treated empirically rather than mechanistically [28], [29]. The emergence of gallium (Ga) and Ga-based alloys has fundamentally altered this situation. These materials melt near or below room temperature, exhibit relatively low toxicity, and can be handled under ambient conditions, enabling systematic investigation of liquid metal interfaces in environments previously inaccessible to molten systems [30], [31], [32]. A defining feature of Ga-based molten metals is their rapid formation of native oxide layers upon exposure to oxygen-containing environments [33]. This oxidation occurs on millisecond timescales and results in the formation of a nanometer-thick solid skin at the liquid surface [34]. This oxide layer profoundly alters interfacial behavior by introducing elastic and plastic mechanical responses to the fluid interface [35], [36]. The resulting structure (a solid skin encapsulating a liquid metallic core) exhibits finite yield stress, supports non-spherical

shapes, and undergoes wrinkling, fracture, and self-healing. These properties are absent in conventional liquids and fundamentally distinguish molten metal interfaces from classical fluid systems [37], [38], [39], [40]. The presence of this oxide skin transforms the molten metal interface into a mechanically active boundary whose behavior depends sensitively on chemical environment, stress state, and electrical conditions [41], [42]. The native oxide layer modulates fluidity and coalescence, imparting mechanical stabilization to molten metal structures, yet remains vulnerable to perturbation via chemical etching, electrochemical reduction, or applied mechanical stress [43], [44]. This dual role (acting both as a barrier and as an enabling element) introduces a new paradigm in interfacial science, where functionality arises from controlled instability rather than static equilibrium [45]. Engineering strategies that manipulate oxide growth, dissolution, and regeneration have therefore become central to controlling molten metal behavior [46], [47], [48].

Furthermore, electrochemical effects differentiate molten metal interfaces from those of insulating or ionic liquids [49], [50], [51]. Because the bulk liquid is electrically conductive, the electrical double layer forms under fundamentally different conditions, with charge redistribution occurring directly within the metallic fluid [52]. This facilitates pronounced electrocapillary phenomena, where modest applied potentials produce significant variations in surface tension, enabling dynamic regulation of droplet morphology, interfacial motion, and force distribution [53]. In addition, interfacial alloying and de-alloying processes provide a pathway for continuous uptake and release of solute ions, turning the interface into a dynamic chemical reservoir rather than a static boundary [54]. The combined mechanical and electrochemical processes enable molten metal interfaces to act as active sites for chemical transformations [55], [56]. In situ reduction reactions, nanoparticle nucleation, and nanomaterial growth can proceed directly at the interface, facilitated by continuous surface renewal and efficient electron transport [57]. Unlike solid catalysts, molten metals are not limited by surface poisoning or irreversible restructuring, as fluidity permits constant reconfiguration of active sites

[58]. This behavior challenges conventional assumptions in catalysis and corrosion science, where solid surfaces are treated as static entities with fixed reactivity [59], [60].

The interplay of surface tension, oxide mechanics, and electrochemical forces also gives rise to complex dynamic behaviors including Marangoni-driven flows, droplet actuation, filament formation, fracture, and re-coalescence [61], [62], [63], [64]. These phenomena support interfacial instabilities and pattern formation that are increasingly exploited in soft robotics, stretchable electronics, and adaptive systems [65], [66]. In these applications, molten metals offer a unique combination of high electrical conductivity, mechanical compliance, and self-healing capability, facilitating functionalities that are challenging or unattainable with conventional solid materials. The significance of molten metal interfaces is increasingly recognized across a wide spectrum of emerging technologies [67]. Liquid metal electrodes are central to high-temperature and room-temperature energy storage systems, while self-healing thermal interface materials rely on fluidity combined with high thermal conductivity [68], [69]. Despite significant progress, a comprehensive and predictive understanding of molten metal interfaces remains elusive [70]. Experimental characterization is complicated by rapid oxidation, coupled mechanical–electrochemical effects, and difficulties in measuring interfacial tension and rheology in the presence of solid skins [71]. Computational approaches, including molecular dynamics and phase-field modeling, have provided valuable insights but often address isolated aspects of behavior [72]. A unified framework that integrates thermodynamics, mechanics, electrochemistry, and kinetics is still lacking [73], [74]. Despite the rapid progress in emerging material platforms such as perovskites, thin films, nanofluids and other advanced functional systems [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], the interfacial science of molten metals remains comparatively underexplored [88]. In particular, a comprehensive framework that integrates their thermodynamics, electrochemical behavior, oxide-mediated mechanics, and dynamic interfacial phenomena is still lacking. Addressing this gap is essential for unlocking the full technological potential of molten metals in next-generation applications.

This review critically addressed the interfacial science of molten metals by examining how liquid-state metallic bonding, thermophysical properties, and rapid oxide formation collectively dictate interfacial structure and dynamics. An overview highlighting the main focus areas and strategic goals of this review is illustrated in [Fig. 1](#). The thermodynamics of molten metal interfaces, the mechanical influence of oxide layers, and key electrochemical phenomena, including electrocapillarity and interfacial alloying, are analyzed to elucidate their impact on interfacial stability, reactivity, and mass transport. Emerging applications and advanced characterization techniques are also discussed in the context of remaining challenges, providing insight into unifying principles and strategies for achieving predictive control over molten metal interfaces.



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Fig. 1. An overview highlighting the main focus areas and strategic goals of this review.

1.1. From metallurgy to soft robotics and catalysis

Molten metals have historically been central to metallurgy, where their primary role was as a process medium for shaping, joining, and alloying solid materials [89]. The evolution of molten metals from traditional metallurgy to modern functional applications is summarized in Table 1, highlighting the exploited material features and emerging roles in

soft robotics, flexible electronics, and catalysis. In these traditional applications, the liquid state was largely considered transient, valuable only for its ability to be cast, molded, or solidified into durable structures. However, contemporary research has revealed that the intrinsic properties of molten metals their high electrical and thermal conductivity, extreme surface tension, fluidity, and rapid surface oxidation can be harnessed far beyond classical metallurgical functions [90]. Today, liquid metals are being integrated into soft robotics as deformable, self-healing conductors and actuators, enabling dynamic reconfiguration and responsive motion not achievable with rigid materials [91]. In catalysis, their liquid surfaces act as adaptive reaction platforms, facilitating in situ nanoparticle growth, alloying, and redox processes under mild or high-temperature conditions [92], [93]. This evolution from passive processing media to active, multifunctional materials illustrates a paradigm shift in the way molten metals are perceived and utilized: from shaping the solid world to engineering dynamic, reconfigurable, and highly functional systems.

Table 1. Evolution of molten metals from traditional metallurgy to the advanced functional applications.

Aspect	Traditional Metallurgy	Functional Applications	Key Material Features Exploited
Primary role	Shaping, casting, joining solids	Active functional component	Fluidity, conductivity, surface activity
State of matter	Transient liquid for processing	Stable or controlled liquid for functionality	Low melting point (Ga alloys), liquid stability
Mechanical function	Solidification determines final shape	Supports deformable, reconfigurable structures	Oxide skin elasticity, yield stress, wrinkling
Electrical function	Passive conductor (limited)	Dynamic conductors for circuits and actuators	High electrical conductivity, electrocapillarity

Aspect	Traditional Metallurgy	Functional Applications	Key Material Features Exploited
Chemical function	Alloying and joining	Catalytic and synthetic platform	Redox activity, in situ nanoparticle formation
Surface behavior	Wetting of molds, adhesion	Adaptive interface for actuation and catalysis	Surface tension, Marangoni flows, oxide-mediated mechanics
Applications	Tools, machinery, structural components	Soft robotics, flexible electronics, catalysis	Self-healing circuits, shape-morphing actuators, reactive surfaces
Scientific challenge	Control of solidification	Control of dynamic interfacial behavior	Coupled mechanics, electrochemistry, and surface reactions

1.2. Fluidity and metallic bonding

The characteristic fluidity of molten metals stems from the nature of metallic bonding, in which valence electrons are freely delocalized across a framework of positively charged ions [94]. Table 2 described key physicochemical consequences of metallic bonding on molten metal fluidity and interfacial behavior. Unlike molecular or covalent liquids, this delocalization allows atoms to move freely relative to one another while preserving strong cohesive forces, ensuing in high density, momentous surface tension, and low viscosity at elevated temperatures [95], [96], [97]. The interplay between atomic mobility and electron delocalization governs both bulk and interfacial properties, including wetting, capillarity, mass transport, and electrochemical behavior [98], [99]. Additionally, the delocalized electron network associated with metallic bonding ensures efficient electronic conduction in the liquid phase, while concurrent surface oxidation generates mechanically robust oxide skins that stabilize non-equilibrium shapes and promote

wrinkling and self-healing behavior [100], [101]. Understanding this interplay is essential for predicting interfacial phenomena in applications such as soft robotics, flexible electronics, catalysis, and energy storage systems.

Table 2. Key physicochemical properties of metallic bonding on molten metal fluidity and interfacial behavior.

Property	Mechanistic Origin	Interfacial Inferences	Applications
Fluidity / viscosity	Delocalized electrons allow atomic mobility	Supports droplet deformation and flow	Soft robotics, reconfigurable circuits
Surface tension	Cohesive metallic interactions	Drives wetting, capillarity, Marangoni flows	Microfluidics, thermal interface control
Electrical conductivity	Free electrons in the liquid	Enables electrocapillarity, surface charge modulation	Actuators, adaptive electronics
Oxide skin development	Surface oxidation of reactive metal	Provides elastic, yield-stress interface	Shape retention, self-healing structures
Thermal conductivity	Electron and phonon transport	Rapid heat dissipation at interfaces	Energy devices, thermal management

1.3. Scope of the review

The rapid expansion of molten metal research in advanced functional applications has revealed that conventional models of liquid interfaces are insufficient to explain their unique behaviors. This review is motivated by the need to critically analyze and unite the fundamental physicochemical principles, interfacial mechanics, electrochemical phenomena, and emerging applications of molten metals, with particular emphasis on the unusual behaviors arising from oxide-mediated elasticity, electrocapillarity, interfacial alloying, and self-healing dynamics. We systematically examine historical

developments, surface and interfacial thermodynamics, confinement effects, and the interplay of mechanics, chemistry, and charge transport, highlighting both established insights and unresolved challenges. The novelty of this work lies in its systematic synthesis of these factors to identify unifying principles, delineate the limitations of current understanding, and propose strategies for predictive design and active control of molten metal interfaces. Unlike previous studies that focus on individual aspects of molten metals, this review provides an integrated and critical perspective that connects experimental observations, in situ characterization techniques, and computational modeling. This study aims to serve as a roadmap for researchers seeking to harness the fluidity, reactivity, and multifunctionality of molten metals in next-generation technologies.

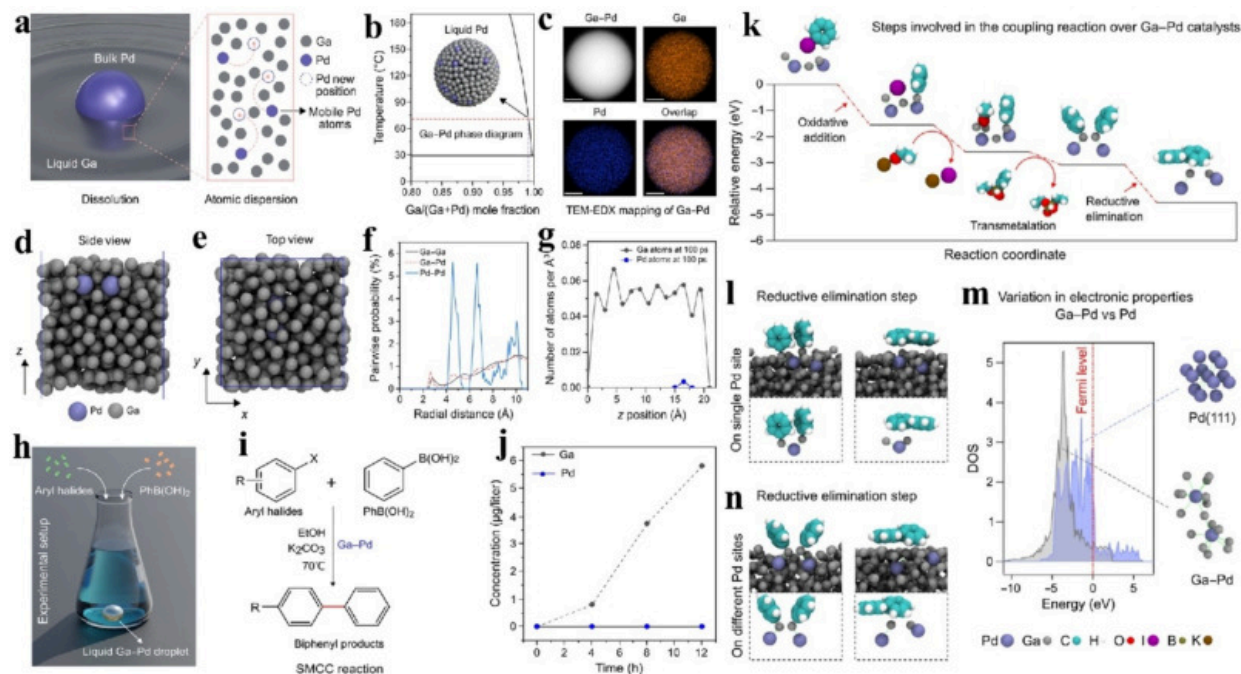
2. Fundamental physicochemical properties of molten metals

Molten metals exhibit a distinctive combination of metallic bonding and fluidity that sets them apart from conventional liquids and solids [102]. In the liquid state, conduction electrons remain delocalized and enabling high electrical and thermal conductivity alongside atomic mobility [103]. This coexistence of electronic coherence and structural disorder governs key physicochemical properties such as high surface tension, low vapor pressure, and strong cohesive forces. Short-range atomic ordering, which is influenced by temperature and alloy composition, plays a critical role in determining viscosity, diffusivity, and interfacial reactivity [104]. Moreover, the thermodynamic tendency of molten metals to minimize surface energy drives spontaneous interactions with surrounding environments, often resulting in rapid oxidation, alloying, or adsorption at interfaces [105]. These fundamental properties collectively define the dynamic behavior of molten metals and underpin their unusual interfacial phenomena, which makes them uniquely responsive, tunable, and functionally active material systems [106].

2.1. Electronic structure and metallic bonding in the liquid state

In the liquid state, molten metals retain the essential characteristics of metallic bonding despite the loss of long-range crystalline order. The valence electrons remain delocalized and form an electron gas that is weakly screened by a dynamically disordered ionic background [107]. This preservation of electronic coherence enables high electrical and thermal conductivity, even as atomic positions fluctuate continuously [108]. Short-range atomic correlations, rather than periodic lattice symmetry, administrate the local electronic structure and lead to temperature- and composition-dependent variations in density of states and transport behavior [109]. Importantly, the fluid nature of the ionic framework allows metallic bonds to continuously break and reform, imparting both mechanical adaptability and rapid electronic response. This duality (electronic rigidity coupled with structural fluidity) distinguishes molten metals from molecular liquids and underlies their exceptional interfacial reactivity, electromechanical tunability, and suitability for dynamic, reconfigurable applications [110], [111], [112]. In 2025, Banna et al, reported that a gallium–palladium (Ga–Pd) catalytic system in which Pd exploits its limited but temperature-dependent solubility in liquid Ga to achieve true atomic dispersion. Pd was dissolved into molten Ga at $\sim 400^\circ\text{C}$ and subsequently cooled to 70°C , where the Pd atoms remain dynamically dispersed within the liquid Ga matrix (Fig. 2a). Phase-diagram and solubility analysis indicated Pd solubilities of ~ 0.005 at % at 50°C and ~ 0.008 at % at 70°C , values that were experimentally confirmed by Inductively coupled plasma–mass spectrometry (ICP–MS) to be ~ 0.01 at % after equilibration at 70°C (Fig. 2b). Minor deviations were attributed to the temperature range used in deriving solubility parameters. Despite employing bulk Ga–Pd droplets during Suzuki–Miyaura C–C coupling (SMCC) reactions, transmission electron microscopy (TEM) coupled with EDX revealed that Pd remained atomically dispersed without clustering (Fig. 2c). Atomic-scale insight was provided by ab initio molecular dynamics simulations, which showed no Pd–Pd aggregation in either the bulk or interfacial regions (Fig. 2d–e). The minimum Pd–Pd separation ($\sim 4.6\text{ \AA}$) was significantly larger than that in metallic Pd, confirming isolation. Consistent with prior observations for Pt and Ni, Pd preferentially resided just beneath the Ga surface rather than directly at the interface. Pair distribution functions and atomic density profiles (Fig. 2f–g) validated these findings, collectively demonstrating a

dynamically stable, atomically dispersed Pd phase in liquid Ga that underpins its unusual catalytic behavior. The catalytic performance of the Ga–Pd system was assessed using a series of model Suzuki–Miyaura C–C coupling reactions, with the batch reaction between iodobenzene and phenylboronic acid selected as the benchmark for optimizing reaction conditions (Fig. 2h–i). Product formation was confirmed by GC–MS and NMR spectroscopy. Under optimized conditions, the Ga–Pd catalyst delivered high yields of biphenyl derivatives across a broad substrate scope, irrespective of electron-donating or electron-withdrawing substituents. While aryl iodides and bromides showed high reactivity, aryl chlorides exhibited only trace conversion, consistent with the higher bond dissociation energy of the C–Cl bond, which limits activation under mild conditions. Catalyst stability was examined by monitoring possible metal leaching during reaction. Periodic sampling followed by ICP–MS analysis (Fig. 2j) revealed negligible variation in Ga concentration and no detectable Pd release, confirming the structural integrity of the Ga–Pd catalyst under operating conditions. Although trace levels of Ga^{3+} (~ 0.8 ppm) were detected, control experiments using an equivalent Ga^{3+} solution showed no catalytic activity, demonstrating that the observed reactivity originates from atomically dispersed Pd within the liquid Ga matrix rather than leached species. Ab initio molecular dynamics (AIMD) simulations were employed to elucidate how Suzuki–Miyaura coupling proceeds at the Ga–Pd liquid metal interface. Several interfacial models were developed to examine the steps of oxidative addition, transmetalation, and reductive elimination. For oxidative addition, the most stable geometry involved the aromatic ring interacting with Ga atoms adjacent to Pd, while the iodine atom coordinated with two Ga atoms located above a neighboring Pd atom (Fig. 2k). Notably, no stable configurations were found where the aromatic ring bonded directly to Pd; in all simulations and geometry optimizations, the ring spontaneously migrated from Pd to nearby Ga atoms, underscoring the dominant catalytic role of the Ga environment rather than exposed Pd sites.



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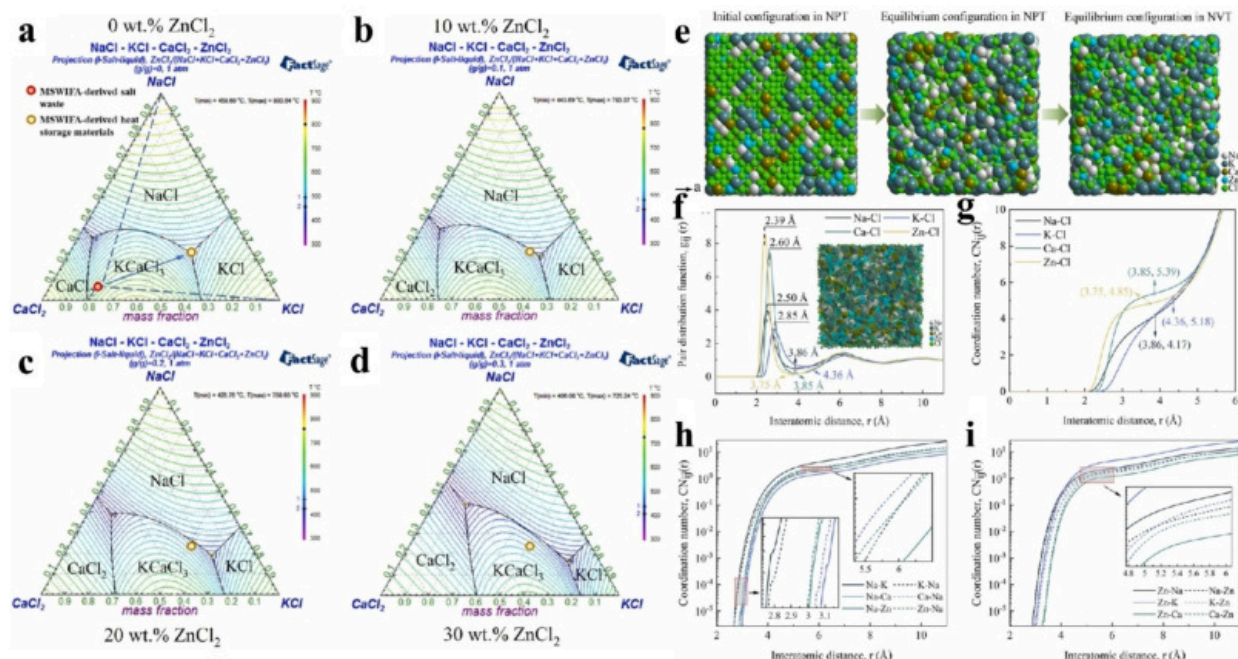
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Fig. 2. (a) Preparation scheme and key features of the Ga-Pd liquid metal system. (b) Ga-Pd phase diagram displaying the temperature-dependent solubility of Pd in liquid Ga. (c) TEM-EDX image. (d-e) Side and top views of AIMD-derived atomic configurations illustrating isolated Pd atoms in the Ga matrix near the interface. (f) Pair distribution function. (g) Atomic density profiles along the surface-normal direction for total atoms and Pd. (h) Experimental setup used for Ga-Pd-catalyzed Suzuki-Miyaura cross-coupling reactions. (i) Representative coupling reaction between aryl halides and phenylboronic acid. (j) Time-dependent ICP-MS analysis. (k) Calculated energy profile of the catalytic cycle. (l-m) AIMD visualizations of reductive elimination occurring via single-Pd and dual-Pd configurations mediated by adjacent Ga atoms. (n) DOS comparison between Ga-Pd and bulk Pd. Reproduced from ref. [113]; Copyrights Science (2025).

The reaction pathway revealed a dynamic interfacial motif in which two Pd atoms reside near the surface, separated by a “bridging” Ga atom and surrounded by several Ga neighbors with average Pd–Ga distances of ~ 2.7 Å. Charge analysis showed that the presence of Pd induces partial positive charges on adjacent Ga atoms, creating an electronically activated Ga cluster at the interface. During oxidative addition, Ga atoms interacting with the C–I bond became more positively charged, while Pd remained largely unchanged. Subsequent ligand exchange with ethoxide and the transmetalation step triggered local rearrangements of Ga atoms and partial charge redistribution, demonstrating the adaptive nature of the liquid interface. During reductive elimination, the system relaxed toward an electronic structure similar to the reactants, while persistent asymmetry highlighted the liquid-state dynamics. Additional interfacial configurations were explored for this step. Unlike solid Pd catalysts, where both aryl groups must coordinate to a single Pd site, AIMD simulations revealed that the aromatic rings in the Ga–Pd system consistently migrated from Pd to adjacent Ga atoms. Two stable motifs emerged: both rings bound to Ga atoms above a single Pd (Fig. 2l) or each ring bound to Ga atoms associated with different Pd atoms (Fig. 2m). Both pathways were energetically favorable, with reductive elimination energies of -2.3 ± 0.4 eV and -1.3 ± 0.7 eV, respectively, while the two-Pd configuration was thermodynamically preferred for the initial state. The unusually high catalytic activity of the Ga–Pd system arises from a combination of electronic synergy and atomic mobility. AIMD-derived diffusion coefficients indicated fast lateral motion of Pd atoms, generating $\sim 1.9 \times 10^5$ new active-site positions per second. This mobility incessantly refreshes the interface, preventing local reactant depletion a limitation intrinsic to solid catalysts. Density of states (DOS) calculations (Fig. 2n) further showed a reduced Pd state density below the Fermi level relative to bulk Pd. Bader charge analysis confirmed electron transfer from Ga to Pd (Pd $\approx -0.72e^-$; Ga $\approx +0.10e^-$), consistent with XPS binding-energy shifts and electronegativity differences. Collectively, these results demonstrate that catalysis in Ga–Pd is governed not by conventional surface Pd sites, but by a dynamically reconfigurable, electronically activated liquid metal interface that enables an effectual low-temperature C–C coupling [113].

2.2. Thermophysical properties relevant to interfaces

The interfacial behavior of molten metals is strongly governed by their thermophysical properties, including density, viscosity, surface tension, thermal conductivity, and specific heat, which dictate flow, wetting, and capillary phenomena at liquid–solid and liquid–gas interfaces. Low viscosity and high surface tension enable rapid spreading and dynamic reshaping of droplets, while thermal conductivity controls heat transfer during reactions or solidification. Temperature-dependent variations in these properties can alter interfacial kinetics, Marangoni flows, and oxide formation, ultimately influencing stability, reactivity, and processability in applications ranging from catalysis to soft robotics. In this context, Chen et al. investigated the NaCl–KCl–CaCl₂ molten salt system with varying ZnCl₂ contents (0–30wt%) to understand structural and interfacial dynamics relevant to metal–salt interactions. Ternary phase diagrams at 0, 10, 20, and 30wt% ZnCl₂, highlighting primary crystal regions and sample points (yellow dots), illustrating how ZnCl₂ modifies phase stability and interfacial environments (Fig. 3a–d). Atomic configurations from molecular simulations (Fig. 3e), while pair distribution functions (PDF) and coordination numbers (CN) of Na–Cl, K–Cl, Ca–Cl, and Zn–Cl, along with polyhedral visualizations highlighting cation-centered chloride coordination (Fig. 3f–g). Short-range ordering of Na and Zn coordination shells show that smaller Na⁺ ions preferentially surround Zn²⁺, reflecting dynamic structural adaptation in the melt (Fig. 3h). Furthermore, CN curves for Zn–cation and cation–Zn pairs (Fig. 3i) reveal that smaller Na⁺ ions preferentially coordinate around Zn²⁺, highlighting short-range structural adaptation in the melt. This local coordination influences viscosity, surface tension, and interfacial behavior, demonstrating how subtle compositional changes modulate thermophysical properties. [114]. Critically, the study demonstrates that even subtle compositional changes can markedly influence local coordination and ionic distributions, which in turn affect viscosity, surface tension, and other thermophysical parameters governing interfacial behavior. This underscores the potential for composition-driven tuning of molten metal and salt interfaces to engineer stable, reactive, and adaptive systems.



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Fig. 3. Ternary phase diagrams of the NaCl–KCl–CaCl₂ system with varying ZnCl₂ content: (a) 0wt%, (b) 10wt%, (c) 20wt%, and (d) 30wt%, showing primary crystal regions (shaded) and experimental sample points (yellow dots). (e) Atomic configurations of simulation models at different compositions. Pair distribution functions (PDF, f) and coordination numbers (CN, g) of Na–Cl, K–Cl, Ca–Cl, and Zn–Cl in sample Z-30, with a polyhedral visualization of cation-centered Cl coordination inset. CN curves for Na–cation and cation–Na (h) and (i) Zn–cation and cation–Zn pairs highlight short-range ordering, where smaller Na⁺ ions dominate the Zn²⁺ coordination shell, reflecting the structural adaptation of the melt at different ZnCl₂ contents. Reproduced from ref. [114]; Copyrights Elsevier (2025).

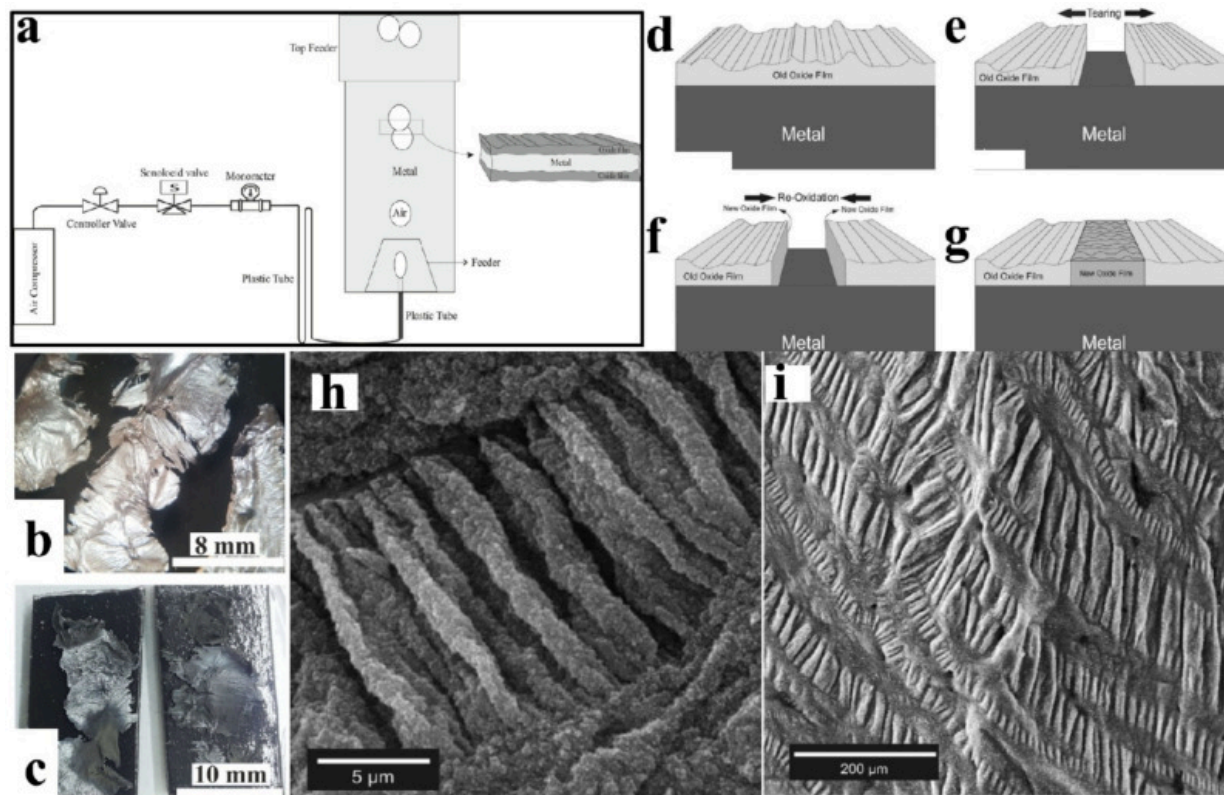
2.3. Oxidation and surface passivation

Molten metals are inherently reactive, and their high surface energy drives rapid oxidation upon exposure to oxygen or other reactive species. This oxidation often forms a self-limiting passivation layer, such as a thin oxide skin, which stabilizes the liquid core, controls wetting, and prevents further bulk degradation. Similarly, alloying or trace-element additions (e.g., Eu in Al–Si alloys) have been shown to modify microstructure and enhance mechanical stability through microstructural spheroidization and refined phase distribution [115]. The thickness, morphology, and mechanical properties of this oxide layer depend on temperature, composition, and alloying elements, influencing interfacial dynamics, droplet reshaping, and capillary-driven flows [116], [117]. Trace additives or alloying elements can tune oxidation kinetics, creating protective skins that are both robust and adaptable, enabling applications in catalysis, soft electronics, and energy devices [118]. Understanding the interplay between oxidation, surface passivation, and molten metal fluidity is crucial for engineering stable, reactive, and programmable liquid interfaces.

In 2022, Jalilvand et al. investigated the surface oxidation behavior of molten Mg–Al alloys using an innovative oxide–metal–oxide (OMO) sandwich method, providing a unique perspective on how molten metal interfaces evolve under reactive conditions [119]. The study employed a controlled bubble generation system to form OMO structures, effectively encapsulating the molten alloy between two oxide layers. This approach allowed the authors to stabilize and directly probe the molten metal interface while simultaneously observing oxidation dynamics, which are critical for understanding interfacial thermophysical behavior and reactivity in molten systems (Fig. 4a).

Macroscopic images of the resulting OMO samples revealed distinct differences between compositions: pure Mg (Fig. 4b) formed relatively uniform oxide layers, whereas the Mg–3Al alloy (Fig. 4c) exhibited more complex surface morphologies, suggesting that Al incorporation modifies oxide growth kinetics and interfacial stability. Intermediate views (Fig. 4d–g) highlighted the layered OMO architecture, showing that the metal core remained sandwiched and protected while the oxide layers developed distinct thicknesses and continuity, emphasizing the role of alloying in controlling surface

dynamics. High-resolution SEM images of the Mg–3Al sandwich (Fig. 4h-i) revealed the characteristic “stitch-line” morphology, where the oxide layers are interconnected through microstructural features at the interface. This stitch-like network indicates a balance between elastic stresses in the oxide layers and the fluidity of the underlying molten metal, illustrating how the interface accommodates thermal expansion, capillary forces, and local compositional variations. The detected microstructural motifs provide direct evidence that interfacial mechanics and thermodynamics govern oxide formation, wetting, and confinement in molten alloys, linking microscopic observations to macroscopic stability. Critically, their work demonstrates that alloying elements such as Al not only modify the chemical reactivity but also influence interfacial mechanical responses, stabilizing the molten metal–oxide interface and providing pathways for controlled oxidation. These observations are directly relevant to understanding molten metal interfaces in energy materials, catalysis, and surface engineering, emphasizing that interfacial structure, composition, and dynamics are intrinsically coupled and can be tailored through composition and processing conditions.



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Fig. 4. (a) Schematic illustration of the bubble generation setup and the formation of the OMO sandwich structure. Macroscopic images of the OMO samples: (b) pure Mg and (c) Mg-3Al. (d-g) Additional views showing the layered OMO architecture. (h-i) SEM images of the Mg-3Al sandwich revealing the detailed interface and microstructural features. Reproduced from ref. [119]; Copyrights Elsevier (2022).

2.4. Critical insights, limitations, and future directions

- Molten metal interfaces are intrinsically dynamic systems governed by thermophysical properties, alloy composition, and interfacial chemistry,

which collectively regulate wetting behavior, capillary forces, and oxide-layer formation.

- Atomic dispersion and mobility of Pd within liquid Ga generate a continuously reconfigurable catalytic interface that facilitates low-temperature Suzuki–Miyaura coupling while mitigating local reactant depletion through rapid interfacial renewal.
- Small compositional variations, such as changes in ZnCl_2 concentration in molten salt environments, significantly alter local coordination structures and ionic distributions, which in turn influence viscosity, surface tension, and interfacial transport phenomena.
- OMO sandwich architectures demonstrate that oxidation in molten alloys often produces structured, self-limiting oxide layers whose continuity and mechanical stability depend strongly on alloying composition and interfacial confinement.
- Despite these advances, several technical bottlenecks remain, including limited mechanistic understanding of liquid-state catalytic interfaces, difficulties in controlling oxide-layer thickness and uniformity, and challenges in maintaining structural stability under prolonged thermal and chemical stress.
- Scalability also represents a major challenge, as many molten metal systems studied at laboratory scale rely on controlled environments or micro-scale configurations that are difficult to translate to industrial processes requiring stable operation, corrosion resistance, and long-term compositional control.
- Addressing these limitations requires integrated experimental, theoretical, and computational approaches capable of resolving interfacial dynamics across atomic, microstructural, and mesoscale regimes.

- Future design strategies should focus on compositionally engineered alloys, controlled oxidation or passivation layers, and dynamically adaptive interfaces that combine catalytic functionality with mechanical and chemical stability.
- Advancing these strategies will enable the rational development of molten metal systems for applications like fluidic technologies.

3. Surface and interfacial thermodynamics

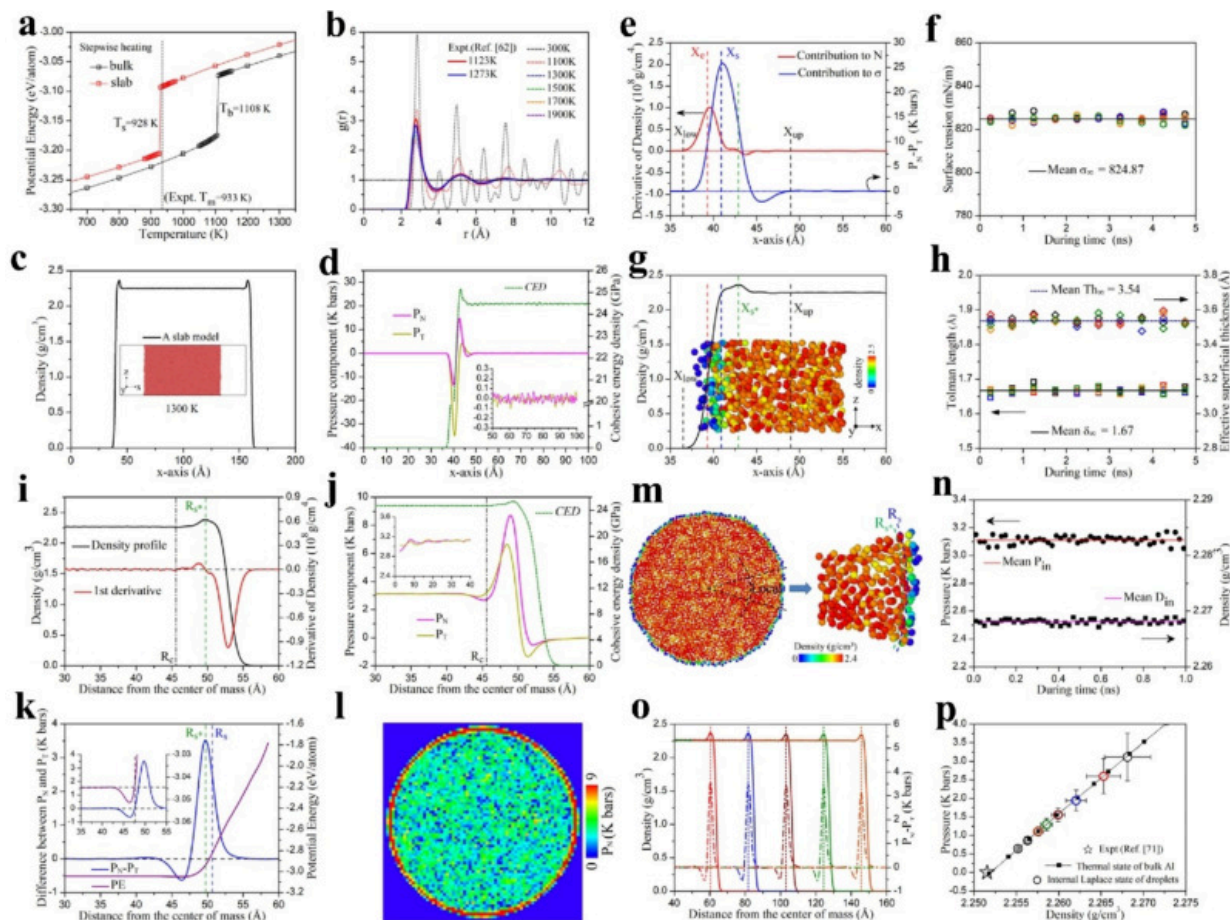
Surface and interfacial thermodynamics govern how molten metals interact with their surroundings and determine their stability, shape, and reactivity [120]. Because molten metals possess high surface energies, they naturally tend to minimize exposed surface area, driving phenomena such as droplet formation, wetting or dewetting, and spontaneous spreading on compatible substrates. The balance among surface tension, interfacial energy, and gravitational or capillary forces decrees interface morphology and dynamic behavior [121]. Temperature, alloy composition, and surrounding atmosphere strongly influence these thermodynamic parameters, leading to changes in wetting angles, interfacial adhesion, and oxide formation. Significantly, small fluctuations in surface energy can drive substantial interfacial reconfiguration, reflecting the dynamic and fluid character of molten metal systems [122]. Understanding these thermodynamic principles is therefore essential for predicting and controlling interfacial stability, chemical reactivity, and functional performance in molten-metal-based systems, ranging from catalysis and metallurgy to soft robotics and adaptive materials.

3.1. Surface energy and wetting behavior

Surface energy is a defining property of molten metals and directly controls their wetting behavior on solid substrates and at liquid–liquid interfaces [123]. Owing to strong metallic cohesion and delocalized electrons, molten metals generally exhibit high surface energies, which favor droplet formation and limited spreading on low-energy surfaces

[124]. Wetting occurs when the interfacial energy between the molten metal and the substrate is sufficiently reduced, either through chemical affinity, alloying, or the presence of surface-active species [125]. Temperature plays a critical role, as increasing thermal energy lowers surface tension and enhances atomic mobility, often improving wetting and interfacial contact. In reactive environments, surface oxidation or adsorption can further modify surface energy, sometimes promoting spreading by forming interfacial layers or, conversely, hindering wetting by creating rigid oxide skins. Consequently, wetting behavior in molten metals emerges from a delicate balance between intrinsic surface energy, interfacial chemistry, and thermal conditions, making it highly tunable but also highly sensitive to processing environments [126], [127]. To elucidate these effects, Chen et al. systematically examined the surface and interfacial thermodynamics of molten aluminum through a combination of atomistic simulations and continuum-scale modeling, offering a comprehensive understanding of the roles of temperature, surface geometry, and curvature in governing wetting characteristics and interfacial stability. The authors further monitored the variation in average atomic potential energy with temperature for both bulk aluminum and slab structures (Fig. 5a). The monotonic increase in potential energy with temperature reflects progressive weakening of atomic cohesion as the solid–liquid transition is approached. The higher energy of the slab relative to bulk highlights the energetic penalty associated with free surfaces, directly linking surface formation to excess surface energy. The radial distribution functions of bulk molten Al at different temperatures (Fig. 5b). The gradual reduction in peak height and positional sharpness with increasing temperature indicates diminishing short-range order, confirming the liquid-state nature while preserving nearest-neighbor metallic correlations. This structural disorder underpins the fluidity and adaptive interfacial behavior of molten aluminum. The ensemble-averaged density profile across a molten Al slab at 1300 K reveals pronounced density oscillations near the surface, followed by a uniform bulk region (Fig. 5c). These oscillations arise from atomic layering induced by surface confinement and are a hallmark of metallic liquids. The inset snapshot visually confirms the formation of a stable planar liquid surface in vacuum. These density variations with mechanical response by showing profiles of the normal pressure,

tangential pressure, and cohesive energy density across the surface (Fig. 5d). The origin of surface tension lies in the imbalance between normal and tangential pressure components at the interface, whereas the lower cohesive energy density reveals the reduced bonding strength of atoms at the surface relative to those in the bulk. Moreover, they defined interfacial boundaries by identifying the surface of tension, equimolar surface, and bulk limits (Fig. 5e). This precise partitioning is essential for quantitatively evaluating surface tension and Tolman length in liquid metals, where the interface is diffuse rather than atomically sharp. An alternative dividing surface introduces, based on the peak in superficial atomic density (Fig. 5f). This approach captures the physically relevant location of maximum surface atom accumulation, emphasizing that different definitions of the interface can yield subtle but meaningful differences in calculated surface properties. The influence of system size (Fig. 5g-h), where surface tension, Tolman length, and effective surface thickness are evaluated for slabs of varying thickness. The results show convergence toward bulk values with increasing slab thickness, confirming that finite-size effects strongly influence interfacial thermodynamics at the nanoscale. Transitioning from planar to curved interfaces, displays the radial density profile of a molten Al nanodroplet (Fig. 5i). Strong density gradients near the droplet surface reveal curvature-enhanced interfacial structuring, which differs fundamentally from planar surfaces. The profiles of normal and tangential pressure components within the droplet demonstrate a curvature-induced pressure imbalance, while the cohesive energy density profile highlights reduced atomic cohesion at the curved interface (Fig. 5j). Difference between normal and tangential pressures directly visualizes together with atomic potential energy, clearly identifying the region responsible for Laplace pressure generation in nanoscale droplets (Fig. 5k). Cross-sectional pressure contour, showing that interfacial stress is highly localized at the droplet surface and becomes increasingly pronounced as droplet size decreases (Fig. 5l).



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Fig. 5. Temperature-dependent structure and surface thermodynamics of molten aluminum. (a) Evolution of average atomic potential energy for bulk Al and Al slabs with temperature. (b) Radial distribution functions of bulk Al at different temperatures, showing liquid-state short-range order. (c) Density profile of a molten Al slab at 1300 K with a snapshot of the simulation model. (d) Normal and tangential pressure components and cohesive energy density across the planar Al surface. (e) Classification of interfacial regions. (f) Surface density profile and partial atomic density contour. (g-h) Dependence

of surface tension, Tolman length, and effective surface thickness on slab thickness. (i–l) Density, pressure, and cohesive energy profiles of a molten Al nanodroplet at 1300 K. (m–o) Atomic density and pressure distributions inside droplets of different radii, identifying homogeneous core regions. (p) Pressure–density relationship of bulk Al compared with the internal Laplace pressure states of molten Al droplets of varying sizes. Reproduced from ref. [128]; Copyright Elsevier (2024).

The atomic-scale manifestation of these effects (Fig. 5m), where density contours illustrate surface roughness and thermal fluctuations in the molten droplet. Enlarged views emphasize the non-uniform nature of curved liquid-metal interfaces. Droplets of different radii are compared, revealing that smaller droplets exhibit sharper density peaks and stronger pressure anisotropy, confirming the dominant role of curvature in nanoscale surface thermodynamics (Fig. 5n). Homogeneous droplet core (Fig. 5o) demonstrating that internal pressure increases with decreasing droplet size, consistent with classical capillarity concepts. Finally, compares the pressure–density relationship of bulk molten Al with the internal states of droplets. The agreement with Laplace pressure predictions confirms that classical thermodynamic principles remain valid even at nanometer scales, provided interfacial thickness and curvature effects are properly accounted for (Fig. 5p). This study convincingly demonstrates that molten aluminum interfaces are governed by a delicate interplay between atomic layering, pressure anisotropy, and surface curvature. This study effectively bridges atomistic-scale features with continuum thermodynamics, thereby validating classical concepts such as surface tension and Laplace pressure at the nanoscale. However, its scope is limited to pure Al under equilibrium conditions, and future work should extend this framework to alloy systems, chemically reactive environments, and dynamically evolving interfaces to better reflect real molten-metal processes [128].

3.2. Interfacial energies with solids, liquids, and gases

As a fundamental thermodynamic parameter, interfacial energy plays a decisive role in determining how molten metals interact with adjacent solid, liquid, and gaseous phases.

It represents the excess free energy associated with creating an interface and directly controls wetting behavior, adhesion, spreading, and interfacial stability. Because molten metals possess high surface energy and strong metallic cohesion, their interfacial energies are typically large and highly sensitive to temperature, composition, and chemical environment [129], [130].

3.2.1. Molten metal–gas interfaces

At molten metal–gas interfaces, interfacial energy is dominated by surface tension, arising from unsatisfied metallic bonds at the liquid surface. These interfaces are highly reactive, particularly in the presence of oxygen or moisture, leading to rapid oxidation and the formation of passivating oxide skins. Such oxide layers can dramatically reduce the effective surface energy, suppress evaporation, and stabilize droplet morphology. Temperature exerts a dual influence on molten metal–gas interfaces: it reduces surface tension by enhancing atomic mobility, while simultaneously accelerating oxidation kinetics. Consequently, these interfaces are inherently dynamic and frequently undergo self-modification [131], [132].

3.2.2. Molten metal–solid interfaces

Interfacial energies between molten metals and solids dictate wetting, infiltration, and bonding, which are critical in casting, brazing, additive manufacturing, and catalysis. Good wetting (low interfacial energy) occurs when chemical affinity, atomic matching, or interfacial reactions promote bonding between the molten metal and the solid substrate. Conversely, when interfacial energies are high, molten metals exhibit poor wetting, resulting in droplet beading and limited adhesion to the substrate. Oxide layers on either phase often act as barriers, increasing interfacial energy and hindering wetting unless disrupted or chemically reduced. Alloying elements and surface treatments are therefore widely used to tune these interfaces [133], [134], [135].

3.2.3. Molten metal–liquid interfaces

Molten metal–liquid interfaces, such as those involving molten salts or immiscible liquid metals, exhibit complex behavior governed by interfacial tension, density contrast, and charge transfer. In molten metal–salt systems, the interplay of partial ionic–metallic interactions and structured interfaces can markedly reduce interfacial energy, promoting stable emulsions and enhanced mass transport. In contrast, immiscible liquid–liquid metal systems maintain high interfacial tension but can still support dynamic phenomena such as Marangoni flows and interfacial diffusion. These interfaces are especially relevant in electrochemical processing, liquid-metal catalysis, and energy storage technologies [136], [137].

3.2.4. Thermodynamic implications

Across all phase combinations, interfacial energy controls equilibrium morphology, interface curvature, and mechanical stability. Representative interfacial energies of molten metals are summarized in Table 3. Minimization of total interfacial free energy drives wetting transitions, droplet spreading, and interface reconstruction. Importantly, interfacial energies are not fixed material constants but evolve with temperature, composition, oxidation state, and external fields, making molten metal interfaces highly tunable yet challenging to control.

Table 3. Representative interfacial energies of molten metals.

Interface type	Typical interfacial energy range	Key factors	Practical implications
Molten metal–gas (vacuum/inert)	0.5–1.5 J m ^{−2}	Metallic bonding strength, temperature	Controls droplet shape, evaporation, capillary flow
Molten metal–gas (oxidizing)	Reduced effective energy (oxide-mediated)	Oxide formation, oxygen activity	Enables surface passivation, droplet stabilization

Interface type	Typical interfacial energy range	Key factors	Practical implications
Molten metal–solid (poor wetting)	$>1.0 \text{ J m}^{-2}$	Chemical incompatibility, oxide barriers	Beading, weak adhesion, limited infiltration
Molten metal–solid (good wetting)	$<0.5 \text{ J m}^{-2}$	Interfacial reactions, alloying, surface chemistry	Strong bonding, spreading, capillary infiltration
Molten metal–molten salt	$0.1\text{--}0.5 \text{ J m}^{-2}$	Ionic–metallic interactions, composition	Enhanced mass transfer, electrochemical activity
Molten metal–liquid metal (immiscible)	$0.5\text{--}2.0 \text{ J m}^{-2}$	Density mismatch, electronic repulsion	Stable phase separation, Marangoni flow

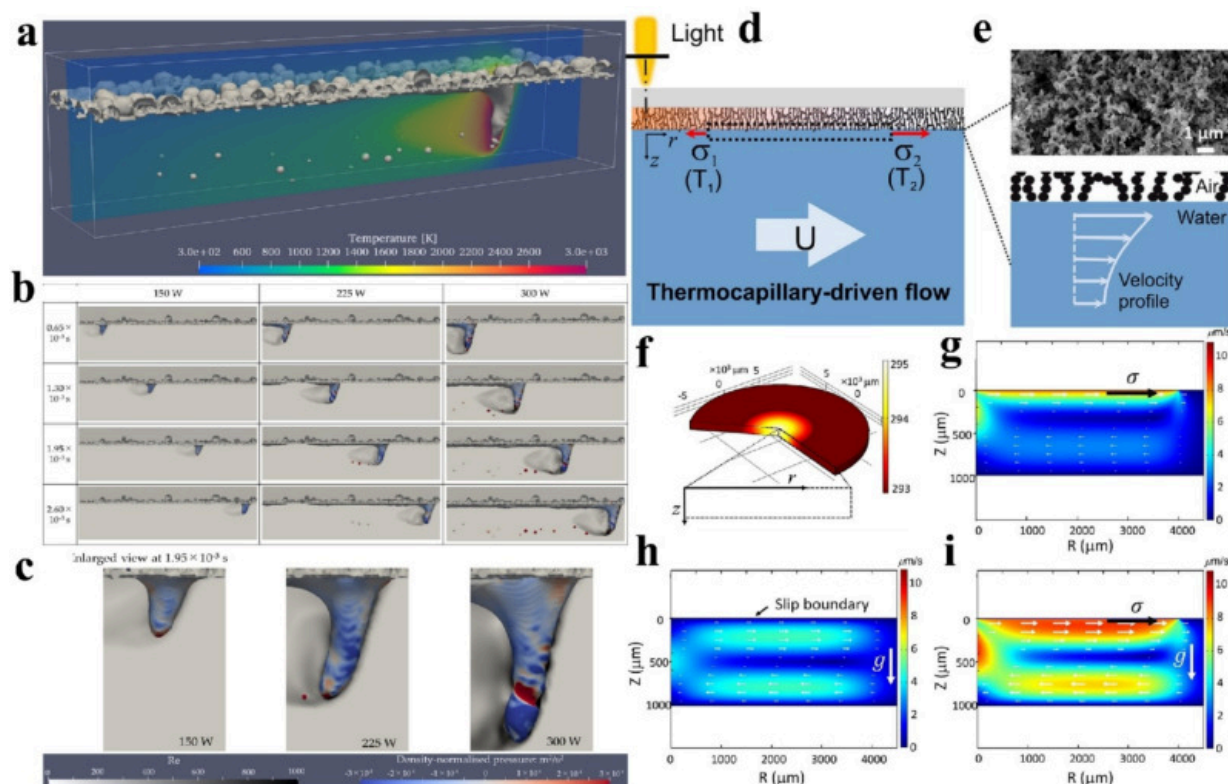
3.3. Capillarity, marangoni effects, and interfacial flows

Representative interfacial energies of capillarity and interfacial flow phenomena play a decisive role in defining the dynamic behavior of molten metals, particularly at small length scales and under thermal or chemical gradients. Owing to their high surface tension and low viscosity, molten metals respond rapidly to interfacial forces, leading to pronounced fluid motion even under modest driving stimuli. These effects govern droplet shape, spreading, stability, and mass transport, and are therefore central to processes ranging from casting and welding to liquid-metal catalysis and soft robotics [138], [139]. Capillary forces arising from surface and interfacial tension drive molten metals to minimize interfacial energy, promoting droplet formation, necking, and efficient infiltration into micro- and nanoscale pores under favorable wetting conditions [140]. At solid interfaces, capillarity dictates whether molten metals spread, bead up, or penetrate capillary channels, directly influencing interfacial contact area and reaction kinetics. The dominance of capillary forces becomes especially evident at the microscale, where gravitational effects are negligible and interfacial energy entirely controls morphology.

Additionally, Marangoni effects originate from gradients in surface tension along an interface, typically induced by temperature differences (thermocapillary effects) or compositional variations (solutocapillary effects). In molten metals, surface tension generally decreases with increasing temperature; thus, thermal gradients generate strong tangential stresses that drive fluid flow from hotter, low-tension regions toward cooler, high-tension regions. Marangoni flows govern heat and mass transport, inducing vigorous internal circulation within droplets or melt pools. In practical systems, these flows critically influence solidification patterns, interfacial mixing, and defect formation. In molten metals, interfacial flows arise from the interplay of capillary pressure, Marangoni stresses, and external forces such as electromagnetic fields or applied shear [141]. The high thermal conductivity of molten metals allows temperature gradients to propagate quickly, sustaining long-range flow patterns across interfaces. At reactive interfaces, such as metal–gas or metal–salt boundaries, interfacial flows continuously reshape the interface, influence oxide growth, and renew reactive sites. In liquid metal catalysts, these flows can prevent local reactant depletion and enhance catalytic turnover by constantly refreshing the active interfacial region. From a broader perspective, capillarity-driven shaping and Marangoni-induced flows endow molten metals with a form of self-organizing behavior, where interfaces dynamically adapt to external conditions. This intrinsic mobility distinguishes molten metals from conventional fluids and solids, enabling reconfigurable interfaces, adaptive wetting, and enhanced transport [142], [143]. Understanding and controlling these interfacial flow phenomena is therefore essential for harnessing molten metals in advanced manufacturing, energy conversion, and responsive material systems.

Katageri et al. provide a clear physical picture of how intense thermal gradients drive complex interfacial flows in molten metals through the coupled action of capillarity and Marangoni stresses [144]. Laser irradiation generates a highly dynamic gas–metal interface characterized by a transient keyhole structure (Fig. 6a). The spatial distribution of the Reynolds number reveals localized regions of vigorous flow, while the pressure contours along the keyhole wall reflect strong recoil pressure and capillary forces acting

simultaneously. These pressure gradients deform the molten metal surface, highlighting how surface tension competes with vapor-induced forces to govern interface stability under extreme thermal conditions. The magnified view of the keyhole captures the early-stage evolution of the molten interface, where rapid heating leads to steep temperature gradients along the liquid surface (Fig. 6b). At this stage, Marangoni stresses arise due to the strong temperature dependence of surface tension, driving tangential flow from hot, low-surface-tension regions toward cooler, high-surface-tension zones. This surface-driven flow redistributes molten metal around the keyhole, directly influencing melt pool shape, penetration depth, and interfacial curvature. The figure underscores that interfacial flow in molten metals is not purely pressure-driven but is critically controlled by thermocapillary forces that act along the interface. Moreover, they conceptually distill this mechanism by illustrating optothermally induced thermocapillary flow. Localized heating produces a nonuniform temperature field at the liquid–gas interface, which translates into a surface tension gradient (Fig. 6c). The resulting Marangoni flow generates a directed surface velocity that couples with bulk convection, even in the absence of strong buoyancy effects. From a theoretical perspective, they demonstrate that capillarity, Marangoni stresses, and pressure-driven flow must be treated as a coupled interfacial phenomenon rather than independent effects. Their study highlights that in molten metal systems subjected to localized energy input, Marangoni-driven interfacial flow can rival or exceed bulk convection in controlling melt dynamics.



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Fig. 6. (a) Snapshot from the simulation showing the gas–metal (liquid or solid) interface under laser irradiation (300 W) at 0.00227 s. (b) Enlarged view of the evolving keyhole at 0.00195 s. (c) Schematic illustrating optothermally induced thermocapillary flow in a liquid contacting a superhydrophobic surface in the Cassie state. Reproduced from ref. [144]; Copyrights Elsevier (2025). (d) Conceptual illustration of thermocapillary flow generated by spatially nonuniform light irradiation on a soot-coated surface, where photothermal heating creates interfacial temperature gradients that drive surface-tension-induced flow. (e) SEM images of the soot microstructure and a schematic of the discontinuous liquid–air interface on the superhydrophobic surface. (f) Finite-element-simulated temperature field under localized heating. Simulated flow fields and velocity

magnitudes for: (g) Marangoni-driven flow without buoyancy, (h) buoyancy-driven flow without Marangoni stress, and (i) combined Marangoni and buoyancy effects. Reproduced from ref. [145]; Copyrights American Chemical Society (2021).

Furthermore, Gao et al. also described the understanding of thermocapillary-driven interfacial flows by systematically decoupling and quantifying the relative roles of Marangoni stresses and buoyancy forces under localized heating conditions [145]. The subsequent figures illustrate how surface tension gradients dominate flow behavior when strong thermal nonuniformities are imposed at liquid interfaces. The conceptual framework of optothermally induced thermocapillary flow at a liquid interface in contact with a superhydrophobic surface (Fig. 6d). Localized light irradiation generates a spatially nonuniform temperature field at the liquid–air interface. This temperature gradient produces a surface tension gradient, which in turn drives interfacial flow from regions of low surface tension (hot zones) toward high surface tension (cool zones). The figure emphasizes that interfacial transport is initiated directly at the surface rather than through bulk convection. SEM images of the soot-based superhydrophobic surface alongside a schematic of the discontinuous liquid–air interface in the Cassie state (Fig. 6e). The trapped air pockets diminish solid–liquid contact, minimizing viscous dissipation and allowing thermocapillary stresses to dominate flow behavior. This structural configuration amplifies interfacial mobility, a condition highly relevant to molten metals where surface-controlled transport often governs flow dynamics. The simulated temperature distribution under localized heating (Fig. 6f). The thermal field is highly localized near the irradiated region, generating steep lateral temperature gradients along the interface. These gradients are directly responsible for generating strong Marangoni stresses, confirming that temperature localization is a key parameter in controlling interfacial flow intensity. Flow field when Marangoni stresses are present without buoyancy effects (Fig. 6g). The velocity vectors reveal strong, surface-confined circulation driven purely by surface tension gradients. This case demonstrates that interfacial flows can be sustained independently of gravity, an especially important insight for molten metals processed at small length scales or under microgravity conditions. The opposite

scenario where buoyancy forces act in the absence of Marangoni stresses was also observed (Fig. 6h). The resulting flow is weaker and more diffused, extending into the bulk rather than remaining localized at the interface. This contrast highlights the limited effectiveness of buoyancy-driven convection compared to thermocapillary transport near interfaces. The coupled case, where both Marangoni stresses and buoyancy forces coexist (Fig. 6i). The results clearly show that surface tension gradients dominate near-interface flow, while buoyancy contributes primarily to bulk circulation. The interfacial velocity magnitude remains governed by thermocapillary forces, confirming their leading role in interfacial transport. Collectively, both studies demonstrate that Marangoni-driven interfacial flow is the primary transport mechanism under localized heating, with buoyancy acting as a secondary effect. For molten metals, which exhibit strong temperature dependence of surface tension and low viscosity, these findings imply that even modest thermal gradients can induce intense interfacial flows. This has direct implications for melt pool stability, defect formation, mixing, and interface reshaping in processes such as laser melting, welding, additive manufacturing, and reactive molten-metal systems.

3.4. Critical insights, limitations, and future directions

- Molten metals possess intrinsically high surface energies, which drive droplet formation, capillary reshaping, and strong sensitivity of interfacial morphology to small thermodynamic perturbations.
- Atomic-scale studies reveal that molten metal surfaces exhibit density oscillations, pressure anisotropy, and reduced cohesive energy, indicating that liquid metal interfaces behave as structured transition regions rather than abrupt boundaries.
- Surface curvature and nanoscale confinement significantly modify interfacial thermodynamics, producing enhanced pressure imbalance and Laplace pressure effects in small droplets.

- Wetting and spreading behavior arise from the coupled influence of surface tension, substrate chemistry, oxidation, and temperature, making molten metal interfaces highly tunable but also environmentally sensitive.
- Capillary and Marangoni forces dominate interfacial transport, particularly under thermal gradients, where thermocapillary flows can exceed buoyancy-driven convection and strongly influence melt pool dynamics.
- Many existing studies focus on pure metals under equilibrium conditions, whereas real systems typically involve multicomponent alloys and reactive environments.
- Quantitative understanding of coupled capillary, Marangoni, and pressure-driven flows remains limited, particularly under strong thermal gradients such as those encountered in laser processing.
- The influence of oxidation, contamination, and transient surface chemistry on interfacial thermodynamics is still difficult to predict and control.
- Current atomistic simulations often face scale limitations, making it challenging to directly bridge nanoscale interfacial structure with macroscopic processing conditions.
- Experimental observation of molten metal interfaces remains challenging due to extreme temperatures and rapid interfacial dynamics, limiting real-time characterization.
- Development of multiscale theoretical models integrating atomistic simulations with continuum thermodynamics to predict molten metal interfacial behavior across length scales.
- Implementation of advanced in situ characterization techniques (e.g., synchrotron imaging, high-temperature microscopy, ultrafast spectroscopy)

to monitor dynamic interfacial evolution.

- Exploration of alloyed and reactive molten metal systems, where composition, surface segregation, and interfacial reactions can be engineered to tailor wetting and transport properties.
- Design of surface-engineered substrates and structured interfaces to control capillary infiltration, spreading, and thermocapillary flow.
- Application of these insights to emerging technologies, including additive manufacturing, liquid-metal catalysis, energy conversion systems, and adaptive materials.

4. Unusual interfacial phenomena in molten metals

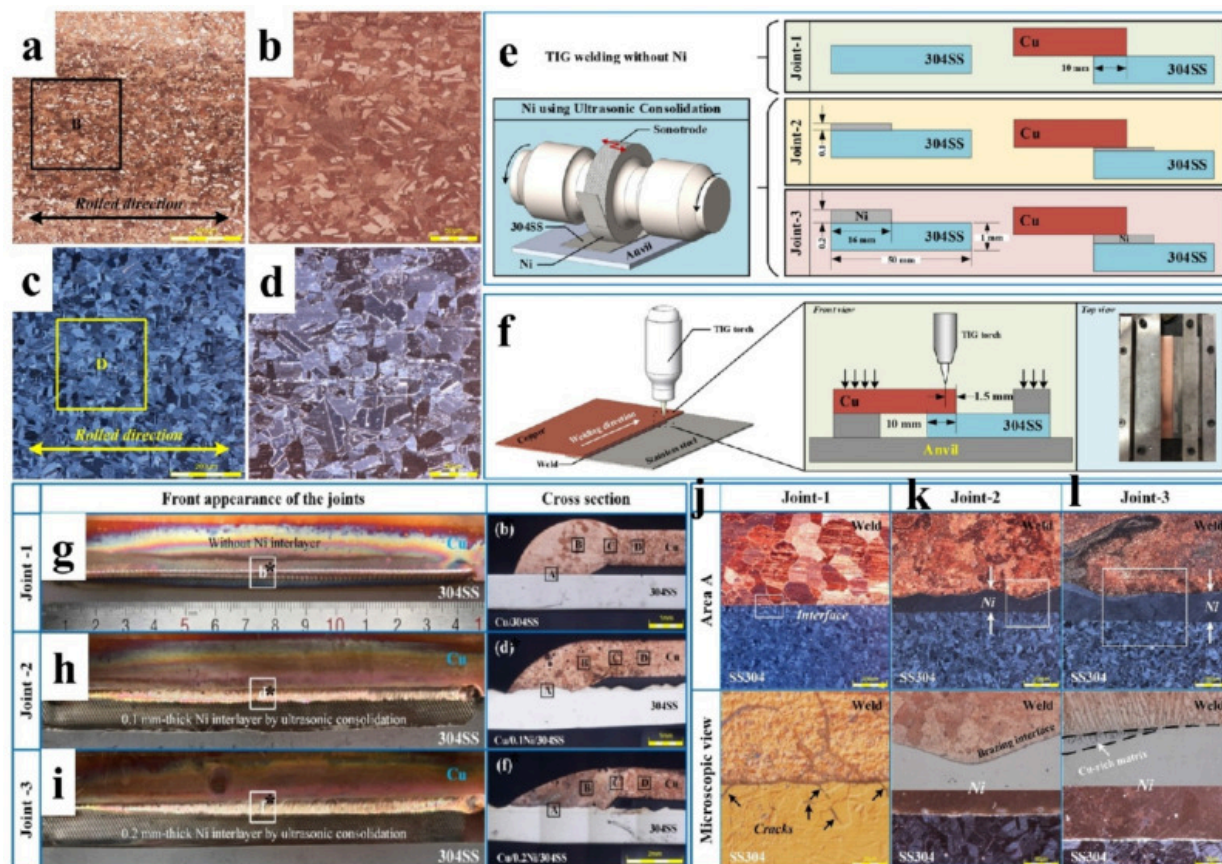
Molten metals exhibit several unusual interfacial phenomena that set them apart from conventional liquids [146]. Their high surface energy and delocalized electrons lead to strong atomic layering at interfaces, producing density oscillations and pressure anisotropy near surfaces. Ultrathin oxide skins can form spontaneously, conferring solid-like elasticity to the liquid surface and enabling nonintuitive behaviors such as shape retention, fracture resistance, and self-healing. Strong temperature or compositional gradients generate intense Marangoni flows that dominate interfacial transport, often overpowering gravity-driven convection. At small length scales, surface curvature significantly alters pressure and cohesion, giving rise to size-dependent surface tension and Laplace pressure effects [147], [148]. Together, these phenomena highlight that molten metal interfaces are highly dynamic, responsive, and governed by a tight coupling between atomic-scale structure and macroscopic fluid behavior [149], [150].

4.1. Liquid–solid transitions at interfaces

Liquid–solid transitions at molten metal interfaces are highly localized and often occur without bulk solidification. Near solid substrates or free surfaces, atomic mobility is reduced, leading to interfacial layering and the emergence of quasi-solid regions. Oxide films or adsorbed species stabilize molten metal interfaces by imparting rigidity, elastic resistance, and yield stress. They localize deformation, support non-equilibrium shapes, and enable self-healing while the underlying metal remains fully liquid. These effects are essential for controlling wetting, patterning, and flow in soft electronics, microfluidics, and additive manufacturing. Thermal gradients and surface curvature strongly influence where and how these transitions occur, with cooling or confinement promoting interfacial freezing [151], [152]. Such localized liquid–solid transitions critically affect wetting, adhesion, interfacial flow, and defect formation, making them central to controlling molten metal behavior in processing and functional applications.

To clarify these effects, Wang et al. carried out a systematic study to understand how liquid–solid interfacial transitions evolve during the joining of dissimilar Cu/304 stainless steel (304SS) systems, with particular emphasis on the role of a Ni interlayer in stabilizing interfacial reactions and microstructure development. Their work directly addresses how molten copper interacts with a solid stainless steel substrate during welding, making it highly relevant to liquid–solid transition phenomena at metallic interfaces [153]. The microstructure of the as-received pure copper exhibits elongated, band-like grains aligned along the rolling direction, indicating a heavily cold-worked state with a strong crystallographic texture (Fig. 7a–b). Such deformation-induced anisotropy is known to influence thermal conductivity and melting behavior, thereby affecting interfacial heat distribution and microstructural evolution during welding. In contrast, the as-received 304 stainless steel shows a relatively equiaxed grain structure with well-defined grain boundaries, reflecting its austenitic microstructure and comparatively lower degree of prior deformation (Fig. 7c–d). This difference in grain morphology and thermophysical properties between copper and 304SS sets the stage for complex liquid–solid interactions when the two materials are joined. The joint configurations investigated by Wang et al. are schematically (Fig. 7e), comparing direct Cu/304SS welding without an interlayer

(Joint-1) to joints incorporating a nickel interlayer of different thicknesses (Joint-2 and Joint-3). The inclusion of Ni is intended to mitigate metallurgical incompatibility by acting as a compositional and thermophysical buffer between the molten copper and solid stainless steel. The tungsten inert gas welding (TIG) welding process employed to fabricate the joints, where localized melting of copper and interfacial reaction with stainless steel governs joint formation (Fig. 7f). Macroscopic appearances and corresponding cross-sectional views of the welded joints reveal a clear influence of the Ni interlayer on weld integrity (Fig. 7g–i). Joint-1 (fabricated in the absence of Ni) displays heterogeneous weld morphology and insufficient interfacial cohesion, indicating kinetically limited wetting and suboptimal metallurgical bonding between molten copper and stainless steel. With the introduction of a Ni interlayer, Joint-2 and Joint-3 exhibit smoother weld surfaces, improved continuity, and more uniform cross-sections, demonstrating enhanced interfacial stability during solidification. Optical metallographic observations further elucidate the evolution of the Cu/304SS interface (Fig. 7j–l). In Joint-1, the interface is relatively sharp and discontinuous, with evidence of interfacial defects and weak metallurgical bonding, consistent with rapid solidification and limited atomic interdiffusion. In contrast, joints containing a Ni interlayer display a more gradual and chemically mediated transition across the interface. The Ni layer promotes mutual solubility and diffusion between Cu and Fe-based phases, leading to improved interfacial cohesion and reduced defect density. Increasing Ni thickness further stabilizes the interface, resulting in a more homogeneous reaction zone and stronger liquid–solid bonding. Overall, this study demonstrates that liquid–solid transitions at Cu/304SS interfaces are highly sensitive to interfacial chemistry and thermal compatibility. The Ni interlayer plays a critical role in regulating melting, diffusion, and solidification at the interface, transforming an otherwise unstable molten copper–solid steel contact into a robust metallurgical joint. These findings highlight the importance of interlayer-assisted interfacial engineering for controlling liquid–solid interactions in dissimilar metal systems.



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Fig. 7. Microstructural characteristics and joint formation in Cu/304SS systems. (a–b) Optical micrographs of the as-received pure copper, showing the rolled grain structure. (c–d) Microstructure of the as-received 304 stainless steel, highlighting its distinct grain morphology. (e) Schematic illustration of the joint configurations, comparing direct Cu/304SS welding without a Ni interlayer (Joint-1) and joints incorporating a Ni interlayer of different thicknesses (Joint-2 and Joint-3). (f) Schematic representation of the TIG welding process used to fabricate the joints. (g–i) Macroscopic appearances and corresponding cross-sectional views of Joint-1, Joint-2, and Joint-3. (j–l) Optical

metallographic images of the Cu/304SS interfacial regions for the three joints.

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4.2. Self-healing and shape reconfiguration

Self-healing and shape reconfiguration are among the most distinctive and scientifically intriguing behaviors of molten metals, arising directly from their fluidity combined with strong metallic cohesion [154]. Unlike solids, molten metals can spontaneously restore continuity after mechanical damage because atomic mobility is high and metallic bonds can rapidly reform [155]. When a molten metal is fractured, cut, or locally disrupted, surface tension immediately drives the liquid to minimize its exposed surface area. This capillary force pulls separated regions back together, allowing the interface to close and the material to “heal” without the need for external intervention. At the atomic scale, delocalized electrons enable rapid rebonding across the rejoined interface, restoring electrical and thermal continuity almost instantaneously [156]. Shape reconfiguration in molten metals is governed by the same thermodynamic tendency to reduce interfacial free energy. High surface tension favors smooth, compact geometries, typically spherical or near-spherical shapes, especially at small length scales where gravity is negligible. When external forces (i.e., stress, pressure, or confinement) are removed, molten metals naturally relax back to energetically favorable configurations. This self-driven reshaping allows molten metals to reversibly change form in response to mechanical deformation, making them fundamentally different from conventional liquids with lower surface energies [157].

A key enabler of controlled self-healing and reconfiguration is the formation of thin surface oxide layers. In many low-melting metals and alloys, such as Ga-based systems, a nanoscale oxide skin forms spontaneously in air. While the interior remains fluid, this oxide layer provides temporary mechanical stability, allowing the molten metal to hold non-spherical shapes or maintain structural features. When the oxide layer is locally ruptured by mechanical stress, chemical reduction, or electrical stimulation the underlying liquid flows and reconnects, effectively healing cracks or restoring continuity.

The oxide skin then reforms, locking the healed structure in place [158]. This dynamic interplay between a fluid core and a solid-like surface layer enables repeated cycles of damage, healing, and reshaping. Thermal and electrical stimuli further enhance reconfigurability. Local heating reduces viscosity and surface tension, accelerating flow and healing rates, while cooling can slow or freeze the structure into a desired shape. Electric fields or currents can induce electromigration or localized Joule heating, selectively triggering flow at targeted regions. These stimuli allow molten metals to actively adapt their shape and connectivity in real time, rather than passively responding to damage.

Healing kinetics are controlled by viscosity, surface tension, and temperature, while the ultimate healed geometry reflects a balance between capillary forces, gravity, and interfacial constraints. Meanwhile, shape reconfiguration arises from dynamic minimization of interfacial free energy, driven by the interplay of capillary, Marangoni, and elastic forces, and is modulated by external fields, surface chemistry, and local stress distributions. These behaviors have profound implications for advanced applications. In soft robotics and stretchable electronics, molten metals can act as self-repairing conductors that maintain electrical functionality even after repeated deformation or fracture. In adaptive materials and reconfigurable devices, their ability to reversibly change shape enables dynamic tuning of mechanical, electrical, and thermal properties [159], [160].

Zhao et al. developed a highly functional self-healing liquid metal composite hydrogel that integrates mechanical adaptability, electrical responsiveness, and magnetic actuation, demonstrating a new paradigm of soft, multifunctional interfaces [161]. The preparation of this composite begins with a carefully designed fabrication process (Fig. 8a), in which a poly(vinyl alcohol) (PVA) polymer network is combined with eutectic gallium–indium–tin (EGaIn–InSn) liquid metal droplets and nickel (Ni) particles to form a dynamic hydrogel. The PVA provides a flexible primary network, while the dispersed liquid metal droplets and Ni particles serve as electrically conductive and magnetically responsive phases. This composition enables a continuous, percolated network of liquid

metal within a viscoelastic matrix, laying the foundation for both self-healing and multifunctional response. The intrinsic self-healing mechanism within the PVA/EGaIn–InSn–Ni hydrogel shown mechanical damage (Fig. 8b). The fluidity of the liquid metal droplets ensures rapid reconnection of conductive pathways, while reversible physical interactions within the PVA network and between liquid metal and polymer chains facilitate mechanical restoration. As the broken interfaces are brought into proximity, surface tension and capillary-driven rearrangements of the liquid metal expedite contact reformation, effectively “healing” the composite without external intervention. The composite’s high-performance schematically highlights electromagnetic interference (EMI) shielding capability (Fig. 8c). The simultaneous presence of liquid metal conductors and heterogeneous interfaces gives rise to multiple shielding mechanisms: conductive loss from the metal network, interfacial polarization due to liquid metal–polymer boundaries, and dipole polarization within the polymer matrix. These coupled mechanisms allow efficient suppression of incident electromagnetic waves over a wide frequency spectrum, rendering the hydrogel suitable for flexible electronics and wearable devices requiring effective EMI shielding. The magnetic responsivity of the composite hydrogel is showing by the embedded Ni particles, enable external magnetic fields to impart controlled motion and manipulation of the hydrogel, demonstrating that the material can be steered, positioned, or reconfigured on demand (Fig. 8d). This magnetic actuation expands the utility of the hydrogel from passive sensing to active, responsive devices. Optical images show its ability to stretch to large strains, exhibit plastic deformation without fracture, and be written on or re-shaped, akin to a soft, conductive putty (Fig. 8e). These deformation modes reflect the unique combination of a flexible polymer matrix with mobile liquid metal inclusions, which together accommodate large strains while maintaining structural integrity. The autonomous self-healing demonstrates capability in practice (Fig. 8f). After being physically cut or damaged, separated hydrogel pieces rapidly reconnect and recover both mechanical and electrical continuity, without external stimuli. This self-recovery behavior is underpinned by the mobility of the liquid metal network and the reversible interactions within the polymeric network. The functional potential of this composite in magnetic actuation and prototyping is also

discussed. The hydrogel is shown completing a circuit under the influence of a magnetic field, proving its utility as a flexible, reconfigurable electrical conductor (Fig. 8g). Magnetic-field-induced shaping demonstrates rapid prototyping capabilities, where the composite responds to magnetic guidance to form predefined geometries (Fig. 8h). This behavior underscores how the combination of liquid metal and magnetic particles enables dynamic reconfiguration beyond static deformation. They shift focus to the electrical sensing functionality of the hydrogel. The relative current change of the material is measured as it is used as a pressure sensor, detecting subtle real-time physiological movements, such as finger bending, wrist bending, and swallowing (Fig. 8i). The continuous variation in current correlates with deformation-induced changes in the percolating liquid metal network, illustrating that mechanical stimuli are transduced into electrical signals with high sensitivity and repeatability. Collectively, Zhao et al. work shows that by integrating liquid metal micro-droplets and magnetic particles into a polymer matrix, one can create a multifunctional soft material capable of self-healing, shape adaptability, EMI shielding, magnetic actuation, and real-time physiological sensing.

composite hydrogel with primary crosslinked networks. (b) Schematic showing the autonomous self-healing mechanism enabled by dynamic interactions within the liquid metal hydrogel. (c) Illustration of the EMI shielding mechanism of the liquid metal hydrogel, highlighting conductive loss, interfacial polarization, and dipole polarization. (d) Schematic demonstrating magnetic-field-driven actuation and movement of the composite hydrogel. (e) Optical photographs showing the flexibility, stretchability, plastic deformation, and writability of the hydrogel. (f) Demonstration of the automatic self-healing behavior after mechanical damage. (g) Magnetic-field-driven movement of the hydrogel in a junction circuit. (h) Magnetic-field-assisted prototyping and shape reconfiguration. (i) Relative current variation of the hydrogel used as a pressure sensor during real-time detection of finger bending, wrist bending, and swallowing. Reproduced from ref. [161]; Copyrights Springer (2023). (j) Application of the hydrogel sandwiched between polyimide films for real-time monitoring of handwritten letters “A”, “B”, and “C”. (k) Schematic illustrating degradation mechanisms of conventional electrodes, including sintering, poisoning, and spoilage. (l) Schematic of the liquid metal electrode in different operational states. (m) Schematic showing the preparation of the Bi_2O_3 –liquid metal mixture. (n) XRD patterns of Bi_2O_3 and the black products collected from the liquid metal surface after cyclic voltammetry. (o) In situ Raman spectra revealing surface chemical transformations of the liquid metal electrode under applied negative potential. (p) In situ XRD patterns tracking the phase evolution and transformation pathway of Bi species during electrohydrodynamic activation. (q) SEM image of Bi collected from the liquid metal surface after electrohydrodynamic evolution. (r) TEM image of the generated crystalline Bi, with the inset showing the corresponding FFT pattern. Reproduced from ref. [162]; Copyrights Nature (2022).

Furthermore, in another study, Hou et al. first outline the deterioration mechanisms of conventional electrodes [162], where active catalysts suffer from structural sintering, surface poisoning, and eventual spoilage during extended electrochemical operation, leading to performance decay and loss of activity (Fig. 8j). The preparation of the Bi_2O_3 –LM composite, designed to exploit electrohydrodynamic evolution for catalytic activation,

is described through an intuitive schematic that illustrates the mixing of bismuth oxide particles into the liquid metal matrix (Fig. 8k). By contrast, they present a liquid metal (LM) electrode paradigm in which the dynamic liquid environment continuously adapts, heals, and reconfigures active sites throughout operation; these schematic captures key states of the LM electrode in action from initial dispersion to evolved reactive surfaces and self-healing restoration after deactivation (Fig. 8l). X-ray diffraction (XRD) analysis reveals the phase changes induced by electrochemical cycling: raw Bi_2O_3 displays characteristic reflections, while the black products collected from the LM surface after cyclic voltammetry show distinct features indicating in situ chemical transformation of Bi species under reactive conditions (Fig. 8m). To track real-time surface evolution, in situ Raman spectroscopy is employed, demonstrating the emergence of new vibrational bands as negative potential is applied, which reflect changes in the chemical state of the LM surface during activation (Fig. 8n). Complementary in situ XRD patterns reveal the pathways and phase transformations of Bi species, confirming that electrohydrodynamic evolution drives the formation of catalytically active Bi phases within the LM environment (Fig. 8o). Morphological characterization of the evolved electrode surface shows that bismuth particles are recovered from the LM matrix and exhibit characteristic features under scanning electron microscopy, indicating the morphology of the active species that participate in electrocatalysis (Fig. 8p). Higher-magnification imaging of these collected Bi regions reveals detailed microstructural organization, highlighting the intimate interaction between deposited Bi and the liquid metal substrate (Fig. 8q). Transmission electron microscopy (TEM) further confirms the crystallinity of the generated bismuth, with lattice fringes and diffraction patterns indicating well-defined crystalline domains formed through electrohydrodynamic evolution (Fig. 8r). Recent studies indicate that, despite the remarkable self-healing, reconfigurability, and interfacial adaptability of molten and liquid metal systems, significant challenges persist in this field. The dynamic nature that enables healing also complicates long-term stability, controllable active-site distribution, and reproducibility under continuous operation. Moreover, current mechanistic understanding of coupled capillary, Marangoni, and electrohydrodynamic effects is still largely qualitative. Issues related to oxidation

sensitivity, material compatibility, scalability, and energy efficiency further limit practical deployment. Addressing these gaps through quantitative in situ characterization and unified interfacial models will be essential for translating these advanced concepts into reliable, real-world technologies.

4.3. Critical insights, limitations, and future directions

- Molten metal interfaces differ fundamentally from conventional liquids due to high surface energy, metallic bonding, and strong coupling between atomic-scale structure and macroscopic transport phenomena, resulting in highly dynamic interfacial behavior.
- Atomic layering, pressure anisotropy, and surface-induced ordering indicate that molten metal interfaces form structured regions where liquid-like mobility coexists with short-range solid-like organization, significantly influencing wetting, adhesion, and mass transport.
- Liquid–solid transitions at interfaces are often localized, governed by substrate chemistry, diffusion pathways, and thermal gradients rather than bulk phase changes.
- Interfacial engineering strategies, such as the introduction of intermediate layers (e.g., Ni in Cu/304SS systems), can regulate diffusion and heat transfer, stabilizing liquid–solid interactions and improving metallurgical bonding.
- Self-healing and shape reconfiguration arise naturally from high atomic mobility and capillary forces, allowing molten metals to restore structural and electrical continuity after mechanical disruption.
- Surface oxide layers introduce temporary mechanical stability while maintaining internal fluidity, enabling phenomena such as shape retention,

crack tolerance, and reversible structural adaptation.

- Incorporating molten metals into polymer or composite matrices translates these interfacial characteristics into multifunctional materials capable of self-healing conductivity, sensing, electromagnetic shielding, and magnetic actuation.
- The highly dynamic nature of molten metal interfaces can lead to instability under prolonged thermal, electrical, or chemical stress, limiting long-term reliability.
- Oxidation, uncontrolled diffusion, and phase segregation may alter interfacial structure over time, affecting reproducibility and performance.
- Interfacial processes involve coupled capillary, Marangoni, electrohydrodynamic, and chemical effects, which remain difficult to quantify and predict.
- Current mechanistic descriptions are often qualitative, and comprehensive multiscale models for molten metal interfacial dynamics are still limited.
- Development of integrated theoretical frameworks combining thermodynamics, interfacial chemistry, and fluid dynamics to better predict molten metal behavior.
- Application of advanced in situ characterization techniques (e.g., high-temperature microscopy, synchrotron spectroscopy, real-time diffraction) to monitor dynamic interfacial evolution.
- Rational interface engineering, including controlled alloying, protective interlayers, and surface chemistry modification, to stabilize molten metal–solid interactions.

- Exploration of hybrid architectures, such as liquid metal–polymer composites and adaptive electrodes, to harness interfacial dynamics for functional materials.
- Translation of laboratory discoveries into energy-efficient processing strategies for applications in metallurgy, soft electronics, catalysis, and energy systems.

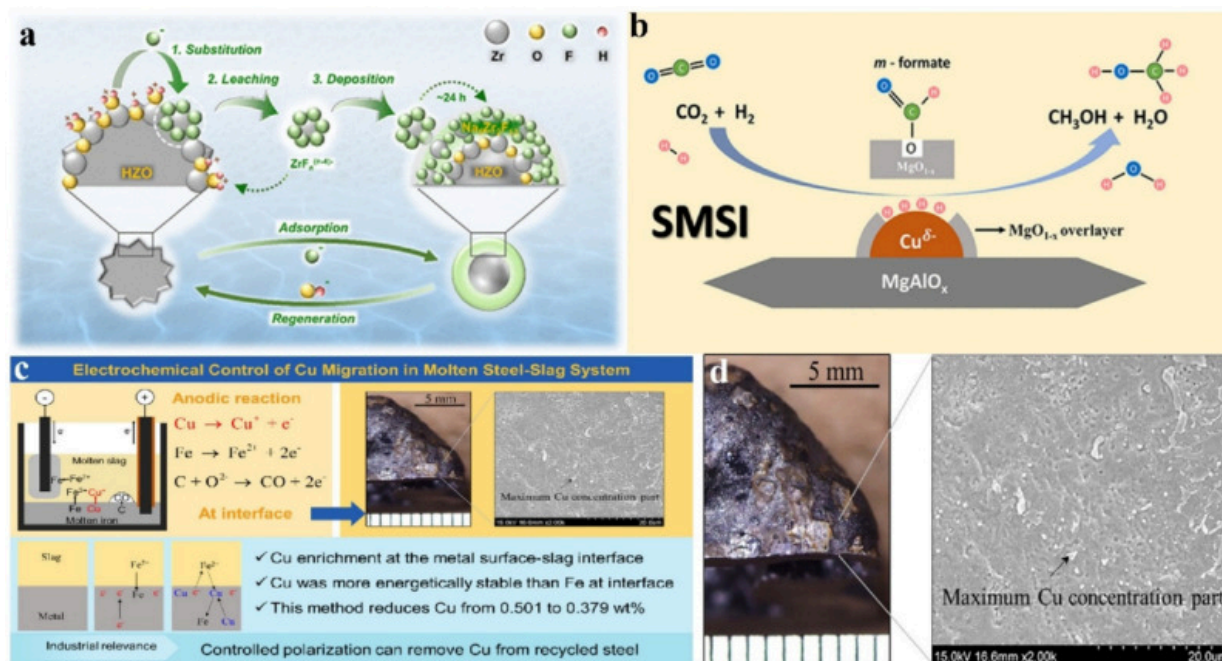
5. Molten metal–oxide interfaces

Molten metal–oxide interfaces are highly complex and dynamic boundaries where thermodynamics, electronic structure, and interfacial mechanics are tightly coupled. At the atomic level, the disparity between delocalized metallic bonding in molten metals and localized ionic or covalent bonding in oxides induces pronounced interfacial reconstruction [163]. Charge transfer and coordination changes create a thin hybrid interfacial zone that is neither purely metallic nor purely oxide-like, yet critically controls interfacial behavior [164]. Thermodynamically, these interfaces are governed by surface and interfacial energies as well as redox-driven chemical potentials. Because oxides typically have lower surface energies than bare molten metals, spontaneous oxide skin formation is common. This ultrathin oxide layer fundamentally alters interfacial energetics, reducing effective surface tension and introducing solid-like elasticity [165]. As a result, oxide-covered molten metals can retain shape, resist flow, and even fracture while remaining liquid beneath. Wetting at molten metal–oxide interfaces is extremely sensitive to chemistry and temperature. Reactive oxides may experience partial reduction or dissolution, enhancing wetting and interfacial adhesion, while inert oxides tend to result in poor wetting and sharp, mechanically weak interfaces. These interfacial behaviors evolve dynamically as oxygen diffuses into the melt and metal migrates into the oxide, continuously altering composition and structure [166]. Mechanically, the oxide skin plays a decisive role by imparting yield stress and elastic response at the surface. Rupture of this skin exposes fresh metal, which rapidly re-oxidizes, enabling a cycle of

flow, healing, and re-passivation. Electronically, differences in work function and electron affinity generate interfacial electric fields that influence adsorption, reaction kinetics, and electrochemical performance, particularly in liquid-metal-based electrodes and catalysts. Molten metal–oxide interfaces are not passive boundaries but active, adaptive regions. Their behavior emerges from the interplay of atomic-scale bonding, mass transport, and surface mechanics, making them central to controlling wetting, stability, self-healing, and functionality in molten and liquid metal systems [167], [168].

Shen et al. used to clearly explain how a molten metal–oxide interface behaves as a dynamic and regenerative system rather than a static boundary [169]. In their model, the oxide surface is shown to continuously evolve through a sequence of coupled chemical and structural processes when exposed to a molten metal environment (Fig. 9a). At the initial stage, substitution occurs when incoming metal species interact with surface cations of the oxide lattice, partially replacing them and locally modifying the coordination environment. This is followed by leaching, where less stable oxide components are selectively dissolved or removed into the surrounding molten phase, creating defect-rich or vacancy-containing regions at the interface. These defects serve as highly active sites that govern subsequent interfacial reactions. Over time, metal atoms or clusters nucleate and grow on the oxide surface, with the preexisting defects providing stabilization for the deposited species. This deposited layer is not permanent; instead, it remains in dynamic equilibrium with the molten environment. Simultaneously, adsorption of reactive species occurs on both the deposited metal and exposed oxide sites, facilitating interfacial reactions such as redox transformations or catalytic steps. After reaction or detachment of deposited species, the oxide surface can partially reconstruct, restoring active sites and enabling repeated reaction cycles. This regenerative behavior explains why molten metal–oxide interfaces often show sustained activity and self-adjusting behavior under harsh thermal and chemical conditions. It is observed that molten metal–oxide interfaces are governed by continuous atomic exchange, defect formation, and surface renewal. They emphasize that this dynamic balance between substitution, dissolution, deposition, and regeneration is central to understanding

interfacial stability, catalytic activity, and long-term performance in molten metal systems. Furthermore, Chen et al. illustrate the role of strong metal–support interaction (SMSI) at a molten metal–oxide interface and how it governs interfacial chemistry during catalytic reactions (Fig. 9b) [170]. The schematic focuses on a Cu–MgAlO_x system, where copper species are in intimate contact with an oxide support under reactive conditions. Chen et al. show that, upon exposure to a reactive atmosphere, Cu species become partially embedded or covered by an ultrathin MgO_x overlayer derived from the MgAlO_x support. This overlayer formation is a hallmark of SMSI and fundamentally alters the electronic environment of Cu. Instead of behaving like bulk metallic copper, the interfacial Cu atoms adopt a Cu^{δ+} state, reflecting charge transfer between the metal and the oxide. This electronic modulation weakens excessive adsorption while stabilizing key intermediates. The figure further depicts how this modified interface directly influences reaction pathways. During CO₂ hydrogenation, CO₂ and H₂ are activated at the Cu–oxide boundary, where the oxide overlayer assists in stabilizing formate (m-formate) intermediates. The cooperative action between Cu sites and the MgO_x overlayer lowers the energy barrier for hydrogenation, ultimately promoting the selective formation of methanol (CH₃OH) while suppressing undesired side reactions. SMSI-induced overlayers act as functional interfacial phases, regulating reactant access, tuning charge distribution, and providing bifunctional sites absent on pristine metals. This dynamic interfacial restructuring enables molten or highly mobile metal systems to adapt under reaction conditions, enhancing catalytic activity, selectivity, and stability. Chen et al. highlight the critical role of interface-engineered electronic states in designing next-generation molten metal and supported metal catalysts.



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Fig. 9. Interfacial phenomena at molten metal–oxide; (a) Schematic illustration of dynamic interfacial processes at an oxide surface, showing substitution, leaching, deposition, adsorption, and regeneration during metal–oxide interaction. Reproduced from ref. [169]; Copyrights American Chemical Society (2025). (b) SMSI mechanism at a Cu/MgAlO_x interface, highlighting oxide overlayer formation and its role in CO_2 hydrogenation pathways. Reproduced from ref. [170]; Copyrights American Chemical Society (2025). (c) Electrochemical control of Cu migration in a molten steel–slag system, depicting anodic interfacial reactions and Cu enrichment/stabilization at the metal–slag interface. (d) Optical and SEM images showing localized regions of maximum Cu concentration at the steel–slag interface after controlled electrochemical polarization. Reproduced from ref. [171]; Copyrights American Chemical Society (2025).

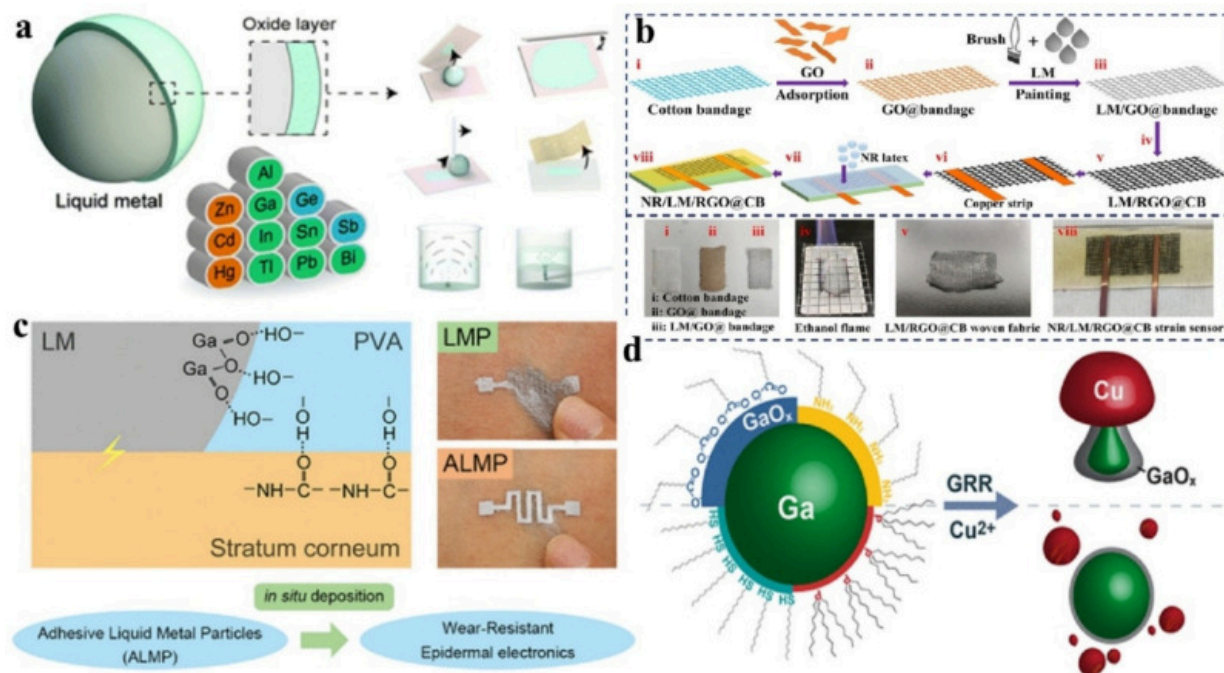
Natsui and coworkers investigate the molten metal–slag (oxide) interface in a steelmaking-relevant system [171] and demonstrate how electrochemical control can actively regulate interfacial reactions and element migration, with a particular focus on copper behavior (Fig. 9c-d). They schematically describe the electrochemical reactions occurring at the molten steel–slag interface under controlled polarization. At the anodic side, copper in the molten steel is oxidized to Cu^+ , while iron simultaneously undergoes oxidation to Fe^{2+} . Carbon oxidation also contributes by generating CO through interfacial reactions. These coupled electrochemical processes establish a potential gradient across the metal–slag interface, which directly drives the redistribution of metallic species. A key outcome highlighted by their study is that Cu becomes energetically more stable at the interface than Fe, leading to preferential Cu enrichment near the boundary. This behavior reflects a delicate balance between thermodynamics and interfacial electrochemistry, where the oxide slag acts not merely as a passive phase but as an active participant in charge and mass transfer. Experimental evidence from macroscopic and microscopic observations reveals localized Cu enrichment at the steel–slag interface. Controlled electrochemical polarization directs Cu from the bulk toward the interface, confirming targeted interfacial redistribution. Significantly, they demonstrate that this interfacial control enables practical removal of Cu from recycled steel, reducing its concentration from approximately 0.501 to 0.379 wt%. This finding is industrially significant, as Cu is a notorious tramp element in steel recycling. From a fundamental perspective, this study reveals that molten metal–oxide interfaces are electrochemically active and tunable, where applied potentials can reshape interfacial stability, element partitioning, and reaction pathways.

5.1. Structure and mechanics of oxide skins

Oxide skins that spontaneously form on molten and liquid metals (particularly Ga-based alloys) are ultrathin (nanometer-scale) yet mechanically robust interfacial layers [172], [173]. Structurally, these skins are amorphous or poorly crystalline metal oxides that tightly adhere to the liquid core. Despite their minimal thickness, they impart solid-like mechanical behavior, enabling shape retention, stretching, and patterning of an otherwise

highly fluid metal [174]. Mechanically, the oxide skin behaves as a yield-stress membrane: it resists deformation under small stresses but fractures or flows when a critical strain is exceeded. This unique combination of fluid interior and elastic–plastic surface underpins the extraordinary reconfigurability, self-healing, and processability of liquid metals in soft and adaptive systems [175].

Aukarasereenont et al. elucidated the pivotal role of the self-formed oxide skin on liquid metals, particularly Ga-based alloys, in governing their mechanical stability and processability (Fig. 10a). They demonstrated that the ultrathin oxide layer acts as a mechanically supportive yet chemically active interface, transforming an otherwise highly fluid metal into a shape-retaining, transferable material. This oxide-enabled viscoelastic behavior allows liquid metals to be patterned, stretched, and conformally transferred onto diverse substrates without external encapsulation. Critically, the study highlights that interfacial oxide formation is not merely a passivation phenomenon but a functional enabler, bridging fluidity with structural integrity (an essential principle for soft electronics and reconfigurable devices) [176]. Furthermore, Lin et al. introduced a scalable strategy for integrating liquid metals with graphene oxide (GO)–based fibrous matrices, yielding multifunctional and mechanically robust textile electronics (Fig. 10b). Their work shows that GO serves as both an adhesion mediator and a diffusion barrier, stabilizing LM coatings during brushing, painting, and lamination processes. The resulting LM/RGO@CB composites exhibit excellent flame resistance, electrical conductivity, and strain sensitivity. From a critical insight, this study advances LM technology from laboratory-scale demonstrations toward wearable and textile-compatible systems, while also exposing challenges related to coating uniformity and long-term mechanical fatigue under repeated deformation [177].



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Fig. 10. (a) Formation of a self-limiting oxide skin on liquid metals enabling shape stability, transfer printing, and patterning. Reproduced from ref. [176]; Copyrights Royal Society of Chemistry (2022). (b) Fabrication of flexible and flame-resistant LM/graphene-oxide-based composite textiles and strain sensors via adsorption, painting, and lamination processes. Reproduced from ref. [177]; Copyrights Elsevier (2024). (c) Interfacial hydrogen-bond-mediated adhesion between LM oxide, polymers, and skin, enabling in situ deposition of adhesive liquid metal particles for wearable, wear-resistant epidermal electronics. Reproduced from ref. [178]; Copyrights American Chemical Society (2022). (d) Ga/GaO_x core-shell interface regulating redox-driven interactions and metal-ion (Cu²⁺) reduction, highlighting interfacial chemistry-controlled functionality. Reproduced from ref. [179]; Copyrights American Chemical Society (2022).

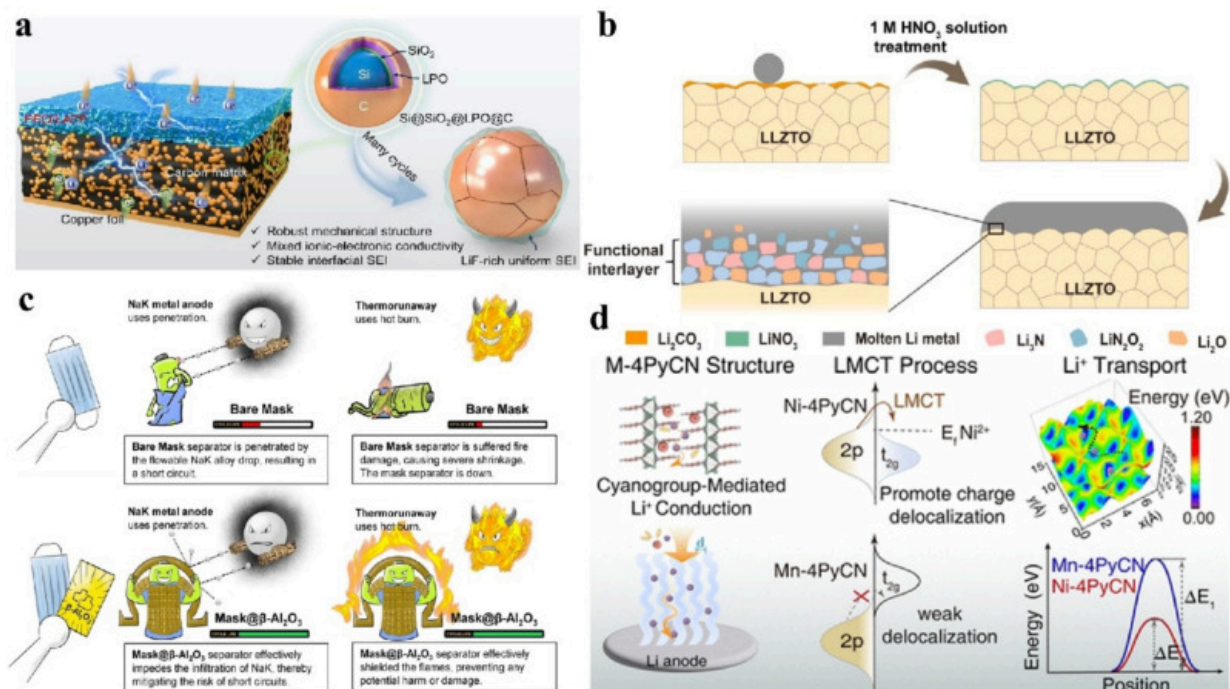
Meanwhile, Ding et al. focused on bio-interfaced liquid metal electronics, revealing how hydrogen-bond-driven interactions between LM oxide layers and polymeric matrices (e.g., PVA) enable strong yet reversible adhesion to human skin (Fig. 10c). They introduced adhesive liquid metal particles (ALMPs) that can be deposited in situ on the epidermis, forming wear-resistant and conformable electronic patterns. Importantly, the study emphasizes molecular-level interfacial chemistry (particularly hydroxyl and amide interactions) as the key to balancing adhesion, comfort, and durability. Critically, this work addresses a long-standing limitation of epidermal electronics by offering a non-invasive, self-adaptive interface, though long-term biocompatibility and oxide stability remain open questions [178].

Additionally, Amorós et al. explored the redox-active Ga/GaO_x interface, demonstrating its ability to mediate spontaneous metal-ion reduction, exemplified by Cu²⁺ to Cu⁰ transformation (Fig. 10d). Their study reveals that the oxide shell is not chemically inert; instead, it functions as a dynamic electrochemical interface capable of driving galvanic replacement and redox reactions. This finding reframes liquid metals as reactive chemical platforms rather than passive conductors. Critically, while this redox activity opens new avenues for catalysis and materials synthesis, it also introduces concerns regarding interfacial instability and abandoned compositional evolution during prolonged operation [179]. Collectively, these studies underscore a unifying theme: the oxide interface is the true functional core of liquid metal systems. Whether enabling mechanical stability, adhesion, bio-integration, or redox activity, interfacial chemistry dictates performance far more than the metallic core itself. Though controlling oxide growth, chemical heterogeneity, and long-term stability remains a critical bottleneck for translating these elegant laboratory concepts into durable, real-world technologies.

5.2. Chemical reactivity and interfacial stability

Molten metals exhibit exceptionally high chemical reactivity due to their fluidic nature, high surface energy, and continuous atomic rearrangement at interfaces [180]. While these features enable self-healing, conformal contact, and adaptive transport, they also

pose serious challenges related to uncontrolled reactions, interfacial degradation, and mechanical instability. Recent studies have advanced the stabilization of molten metal–solid interfaces under extreme electrochemical conditions by precisely tuning interfacial chemistry, introducing protective oxide or polymer layers, and modulating electronic structures to control charge transfer, adhesion, and reactivity at the interface. Gu et al. demonstrated that stabilizing molten lithium at the electrode interface [181] requires simultaneous control of mechanical integrity and chemical reactivity (Fig. 11a). Their work introduced a carbon-based composite host on a copper current collector, which homogenizes lithium deposition and buffers volume fluctuations. More importantly, the host promotes the formation of a LiF-rich solid electrolyte interphase (SEI), which is chemically stable against molten lithium and suppresses parasitic reactions. By combining mixed ionic–electronic conductivity with a mechanically robust framework, Gu et al. showed that molten lithium can be confined and stabilized rather than aggressively reacting with its surroundings. In a complementary approach, Kong et al. focused on stabilizing the highly reactive interface between molten lithium metal and ceramic solid electrolytes (Fig. 11b) [182]. They revealed that untreated LLZTO suffers from poor wettability and interfacial decomposition when in contact with molten lithium. Through controlled acid treatment, a functional interlayer rich in Li_2CO_3 , Li_3N , and LiNO_3 was formed, which acts as a chemically compatible buffer. This interlayer reduces interfacial resistance, facilitates Li^+ transport, and suppresses direct chemical attack by molten lithium, highlighting that interfacial chemistry (not bulk stability) governs long-term compatibility. In 2024, Chen et al. reported the molten metal reactivity to NaK liquid metal anodes, which is even more chemically aggressive than lithium (Fig. 11c). Their study clearly illustrated the catastrophic consequences of uncontrolled reactivity: bare polymer separators were easily penetrated by liquid NaK, leading to short circuits and thermal runaway. By introducing a $\beta\text{-Al}_2\text{O}_3$ ceramic-modified separator, they effectively blocked liquid metal infiltration while maintaining ion transport. This work emphasizes that physical confinement coupled with chemical inertness is essential for managing molten metal reactivity in high-risk battery systems [183].



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Fig. 11. (a) Schematic illustration of a composite lithium anode architecture featuring a carbon matrix on Cu foil, enabling uniform Li plating, mixed ionic–electronic transport, and a stable LiF-rich SEI during long-term cycling. Reproduced from ref. [181]; Copyrights Elsevier (2022). (b) Formation of a functional interlayer on LLZTO solid electrolyte via acid treatment, improving interfacial contact, Li⁺ transport, and compatibility with molten Li metal. Reproduced from ref. [182]; Copyrights Elsevier (2026) (c) Comparison of bare and β -Al₂O₃-modified separators in NaK metal batteries, highlighting the protective role of ceramic coatings against metal penetration, short circuiting, and thermal runaway. Reproduced from ref. [183]; Copyrights Elsevier (2024). (d) Molecular-level design of M-4PyCN interlayers (Ni vs Mn) illustrating cyano-mediated Li⁺ conduction, LMCT, and enhanced charge delocalization for stabilized Li metal interfaces. Reproduced from ref. [184]; Copyrights American Chemical Society (2026).

At the molecular scale, Jiang et al. provided a fundamentally different strategy by tailoring electronic interactions at the molten metal interface (Fig. 11d). Using cyano-group-coordinated metal-organic interlayers (M-4PyCN), they showed that ligand-to-metal charge transfer (LMCT) can delocalize electrons at the interface, reducing localized charge accumulation that typically accelerates side reactions. The Ni-4PyCN system, in particular, enhanced Li^+ conduction while stabilizing the molten lithium surface electronically, demonstrating that interfacial electronic structure engineering can be as powerful as physical barriers in controlling molten metal reactivity [184]. These studies reveal that chemical reactivity of molten metals is not an inherent limitation but a controllable parameter. Stability emerges when molten metals are guided by synergistic strategies involving protective interphases, ceramic barriers, conductive hosts, and electronic delocalization. The key challenge ahead lies in integrating these approaches into scalable systems where molten metals can operate safely under high current densities and long-term cycling without sacrificing their unique self-healing and adaptive advantages.

5.3. Critical insights, limitations, and future directions

- Molten metal-oxide interfaces behave as dynamic and regenerative systems, where substitution, leaching, deposition, and adsorption processes continuously reconstruct the interfacial structure.
- The formation of ultrathin oxide skins fundamentally transforms molten metals by imparting solid-like mechanical properties, enabling shape stability, patterning, and self-healing behavior.
- SMSI play a decisive role in tuning interfacial electronic structure, modifying adsorption energies, and improving catalytic selectivity.
- Interfacial regions often exhibit hybrid electronic states (e.g., partially oxidized metal species), demonstrating that catalytic and electrochemical

activity primarily originates from the metal–oxide boundary rather than the bulk phases.

- Electrochemical polarization can actively manipulate element migration and interfacial composition, revealing that molten metal–oxide boundaries can function as controllable reaction zones in metallurgical and catalytic systems.
- Oxide skins and engineered interphases can serve dual roles as both protective barriers and reactive interfaces, enabling stabilization of highly reactive molten metals in batteries and energy systems.
- Despite extensive experimental observations, the atomic-scale structure of molten metal–oxide interfaces remains poorly resolved, particularly under high-temperature and reactive environments.
- Oxide skins often exhibit chemical heterogeneity and structural instability, which can lead to unpredictable mechanical behavior and interfacial degradation over long operation times.
- Many current studies rely on idealized laboratory conditions, whereas real-world environments involve complex atmospheres, impurities, and dynamic stresses that significantly alter interfacial behavior.
- The dynamic evolution of oxide layers during repeated oxidation–reduction cycles is not fully understood, limiting precise control of interfacial properties.
- Strategies for stabilizing molten metal interfaces (e.g., protective layers or ceramic barriers) may introduce additional resistance to mass or charge transport, potentially compromising system performance.

- Large-scale industrial implementation remains challenging due to difficulties in controlling oxide growth, defect chemistry, and interfacial uniformity.
- Development of in situ and operando characterization techniques capable of probing molten metal–oxide interfaces under realistic thermal and chemical conditions will be critical for resolving their atomic-scale dynamics.
- Advanced multiscale computational modeling, integrating thermodynamics, electronic structure, and mass transport, should be employed to predict interfacial reconstruction and guide rational interface design.
- Future research should focus on precise engineering of oxide skins and interlayers, enabling controlled electronic states, defect structures, and chemical reactivity.
- Designing self-regulating interfaces that dynamically adjust oxidation state, thickness, and composition could significantly enhance stability in molten metal batteries and catalytic systems.
- Exploration of hybrid interfaces combining liquid metals with 2D materials, polymers, or porous frameworks may open new possibilities for flexible electronics, catalysis, and energy storage technologies.
- Finally, translating laboratory insights into practical applications will require scalable fabrication strategies and long-term stability studies under industrially relevant operating conditions.

6. Molten metal interfaces in confined and soft matter systems

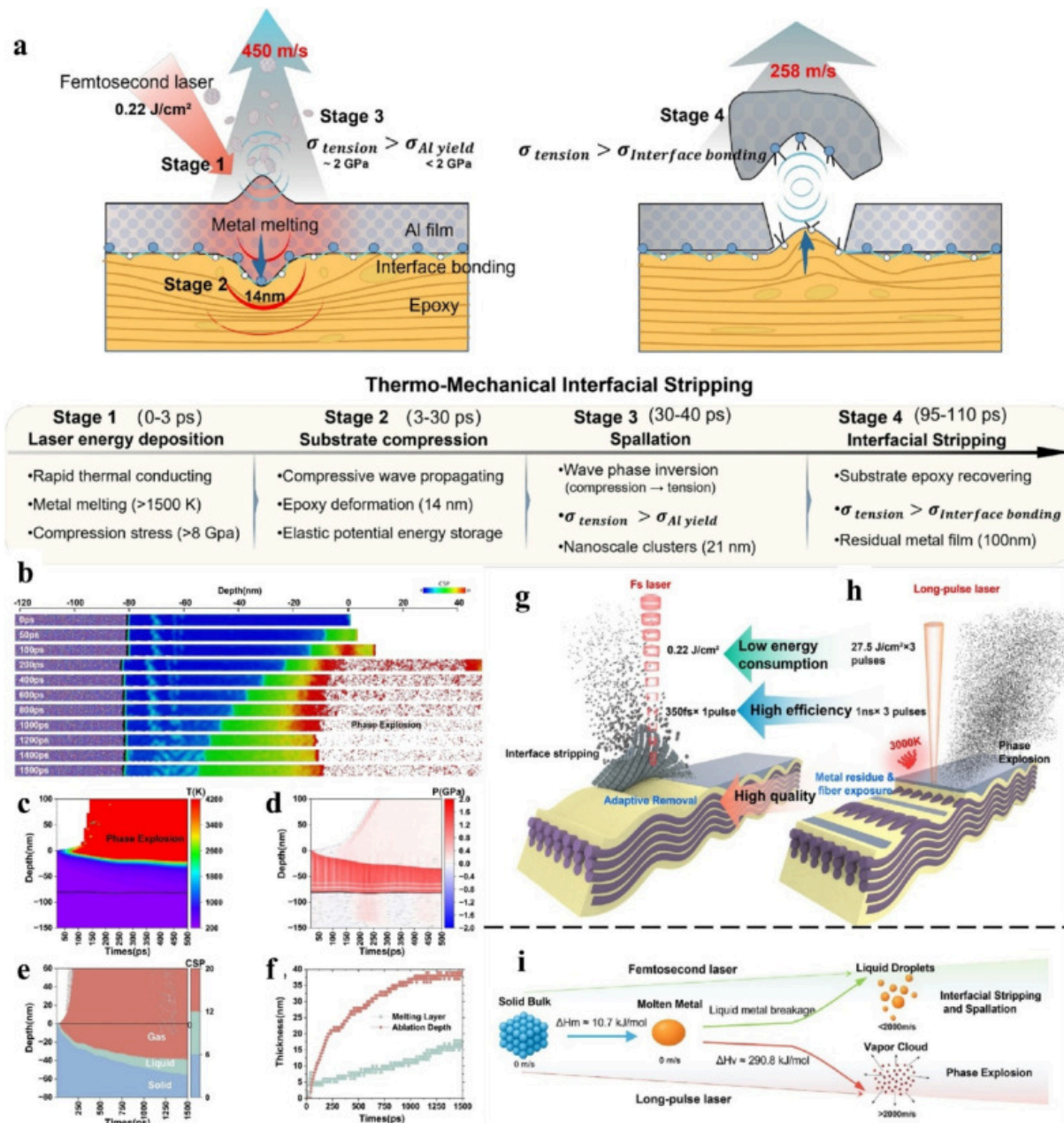
Molten metal interfaces in confined and soft matter systems exhibit behaviors that differ fundamentally from those in bulk environments due to geometric restriction, interfacial forces, and mechanical compliance of the surrounding medium [185]. Micro- and nanoscale confinement suppresses turbulence and amplifies capillary and surface tension effects, stabilizing molten metals in channels, pores, droplets, or thin films despite intrinsically high surface energy [186]. In soft matrices such as polymers, hydrogels, elastomers, and biological tissues, the liquid metal interface becomes mechanically coupled to a deformable host, enabling reversible shape change, stretchability, and self-healing without loss of electrical continuity. The presence of an ultrathin oxide skin further stabilizes these confined interfaces by introducing yield stress and elastic resistance, allowing molten metals to maintain non-spherical geometries under stress. Chemically, confinement limits mass transport and slows oxidation or dissolution, while soft matter interfaces promote adhesion through hydrogen bonding, van der Waals interactions, or interfacial wetting mediated by the oxide layer [187], [188]. Collectively, these effects transform molten metals from highly reactive fluids into adaptive, mechanically integrated interfacial materials, underpinning their use in soft electronics, reconfigurable circuits, biomedical devices, and responsive composite systems [189], [190].

6.1. Micro- and nanoscale confinement

Micro- and nanoscale confinement profoundly alters the behavior of molten metal interfaces by shifting dominance from bulk fluid dynamics to surface- and interface-governed phenomena [191]. When confined within microchannels, nanopores, droplets, or thin interfacial layers, molten metals experience enhanced capillary forces and suppressed gravitational and inertial effects, allowing otherwise unstable liquid metals to be immobilized or precisely guided. At these length scales, surface tension and oxide-skin elasticity dictate shape, flow, and stability, enabling the formation of wires, films, and patterned features with long-term structural integrity. Confinement also restricts atomic diffusion and limits oxygen access, slowing interfacial reactions and promoting metastable configurations. Furthermore, size effects and lattice distortion in nanoscale

alloys can influence molten metal behavior under confined conditions [192]. As a result, molten metals under micro- and nanoscale confinement behave less like free-flowing liquids and more like adaptive, surface-stabilized materials, which is critical for applications in microelectronics, microfluidics, catalysis, and nanoscale energy systems [193].

To clarify it, Huang et al. (2026) systematically elucidated how interfacial stripping governs laser-induced removal of confined metallic films, combining Temperature model–molecular dynamics (TTM–MD) simulations with mechanistic analysis to move beyond the conventional evaporation-dominated view of laser ablation [194]. They demonstrated that, under intense laser irradiation, molten metal behavior is dictated by interfacial thermomechanical instability rather than surface vaporization. They first introduce the interfacial stripping removal mechanism, where rapid laser heating drives the metal film into a molten and highly pressurized state, while the underlying interface acts as a mechanical weak plane. The mismatch in thermal expansion and stress response across this interface generates strong tensile forces, causing the molten layer to detach and be stripped away as a whole rather than eroding atom by atom (Fig. 12a). Atomistic insight into this process is provided through TTM–MD simulations of an 80 nm Al film irradiated by a single 1 ns laser pulse at 27.5 J cm^{-2} . The atomic snapshots reveal uniform and rapid melting throughout the film thickness, followed by collective upward motion of molten atoms initiated near the interface, directly visualizing the onset of interfacial stripping instead of gradual surface recession (Fig. 12b).



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Fig. 12. (a) Schematic illustration of the interfacial stripping mechanism, highlighting how laser–matter interaction induces material removal at the interface. (b) Atomic-scale snapshots from TTM–MD simulations. (c–d) Spatiotemporal evolution of lattice temperature and pressure during irradiation, revealing the coupled thermal–mechanical response of the film. (e–f) Time-dependent progression of phase transformation, melting, and ablation depth in Al. (g–h) Schematic comparison of material processing mechanisms under fs-laser irradiation versus long-pulse laser irradiation, emphasizing the distinct interfacial stripping behavior enabled by fs lasers. (i) Comparative analysis of energy consumption, demonstrating the efficiency advantages of fs laser processing. Reproduced from ref. [194]; Copyrights Elsevier (2026).

The driving forces behind this response are clarified by analyzing the spatiotemporal evolution of lattice temperature and pressure. The simulations show ultrafast lattice heating to well above the melting point, accompanied by the buildup of extreme compressive stress that subsequently relaxes into a tensile stress pulse at the interface. This tensile stress is identified as the critical trigger for interfacial separation and material removal (Fig. 12c–d). The temporal evolution of phase transitions further confirms this mechanism. Melting occurs almost instantaneously across the entire film, while ablation depth increases abruptly once interfacial tensile stress exceeds the cohesive strength of molten Al. This step-like ablation behavior highlights that material removal is controlled by stress-wave–induced detachment rather than continuous thermal evaporation (Fig. 12e–f). Then compare femtosecond and long-pulse laser processing to illustrate the advantages of interfacial stripping–dominated removal. Femtosecond laser irradiation confines energy deposition within ultrashort timescales, maximizing stress localization at the interface and enabling precise stripping with minimal thermal damage, whereas long-pulse irradiation allows heat diffusion to dissipate stress, reducing stripping efficiency and increasing collateral damage (Fig. 12g–h). Finally, an energy consumption analysis highlights the practical significance of the interfacial stripping mechanism, showing that femtosecond laser processing achieves efficient material removal at

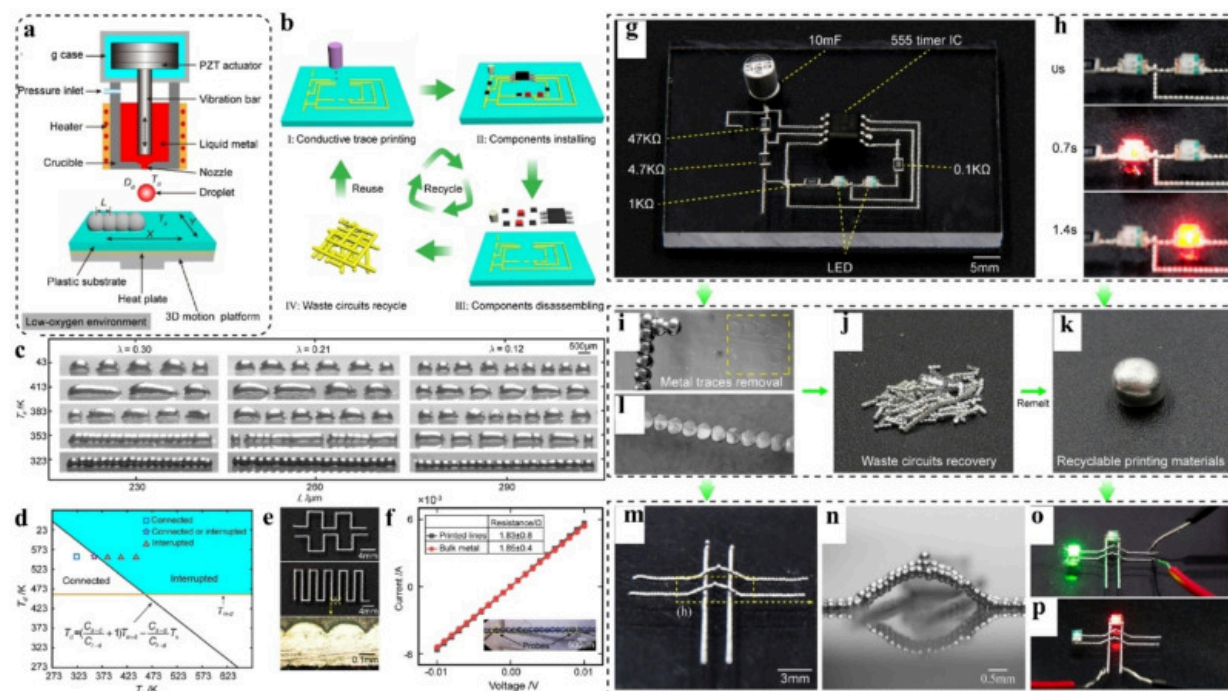
substantially lower energy input by exploiting interfacial stress-driven detachment rather than energetically costly vaporization (Fig. 12i).

6.2. Interaction with soft substrates and polymers

The interaction of molten metals with soft substrates and polymers represents a unique interfacial regime where fluid metallic behavior meets viscoelastic, thermally responsive matter [195]. Unlike rigid inorganic solids, polymers can locally deform, soften, or flow when exposed to the heat and surface forces of molten metals. This mutual adaptability allows molten metals to form intimate, conformal interfaces that are difficult to achieve with conventional solid metals. At the core of this interaction is thermal coupling. When a molten metal droplet contacts a polymer surface, heat transfer occurs rapidly, often bringing the polymer above its glass transition or melting temperature in a highly localized region [196]. Transient softening allows the molten metal to partially penetrate the polymer substrate, forming a mechanically interlocked interface in addition to chemical adhesion. Upon cooling, the polymer resolidifies, securing the metal while maintaining overall flexibility. Interfacial behavior is strongly governed by surface chemistry and oxide layers, resulting in hybrid soft–liquid mechanical characteristics [197], [198]. The polymer substrate absorbs strain and redistributes stress, while the molten metal maintains electrical continuity through its fluid core. Under deformation, the metal can elongate, reconnect, or self-heal, while the polymer provides structural support and fatigue resistance. This synergy enables extreme stretchability, foldability, and resilience under repeated mechanical cycling [199]. From a functional standpoint, molten metal–polymer interfaces enable reconfigurable and recyclable electronics. Because the interaction is largely physical rather than covalent, metal traces can be detached, remelted, and reused without damaging the substrate. At the same time, the interface remains sufficiently robust for stable device operation, including sensing, actuation, and signal transmission. Critically, the stability of these systems depends on controlling interfacial energy, oxidation, and thermal history. Excessive oxidation can embrittle the interface, while insufficient oxide formation leads to poor adhesion.

Similarly, mismatches in thermal expansion or repeated heating–cooling cycles can gradually degrade interfacial integrity.

Duo et al. reported a systematic and practical strategy for fabricating recyclable electronic circuits using a drop-on-demand printing approach, in which molten metal droplets are partially embedded into thermoplastic substrates [200]. Their study demonstrates how precise control over droplet dynamics, interfacial wetting, and thermal conditions enables reliable circuit formation while preserving the ability to fully recover and reuse the metal. The process is built around a metal-droplet-based 3D printing platform, where individual molten droplets are generated and accurately deposited onto plastic substrates (Fig. 13a). Upon impact, the thermal energy of the droplet locally softens the polymer, allowing the metal to partially sink into the surface. This creates a mechanically anchored yet non-permanent metal–polymer interface, eliminating the need for adhesives or chemical surface treatments. The overall fabrication concept is illustrated through a schematic showing the printing, operation, and recycling cycle (Fig. 13b). Conductive traces are printed directly onto plastic substrates, used as functional circuits, and later removed and recycled. This closed-loop process highlights the sustainability of the technique and its potential to reduce electronic waste. The formation of continuous conductive lines is shown to depend strongly on the printing step length (L) and the substrate temperature (T_s) (Fig. 13c). Shorter step lengths combined with elevated substrate temperatures promote droplet coalescence, resulting in smooth and electrically continuous traces. This behavior is further quantified by plotting droplet temperature (T_d) versus substrate temperature, which defines clear processing windows for achieving connected lines (Fig. 13d). These parameter maps serve as practical guidelines for stable printing.



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Fig. 13. (a) Conceptual illustration of a metal-droplet-based 3D printing platform designed for direct circuit fabrication. (b) Schematic showing the step-by-step process used to print and later recycle conductive circuits on plastic substrates using controlled metal droplet ejection. (c) Optical images of printed conductive lines formed by adjusting the printing step length and T_s . (d) Relationship between droplet T_d and T_s , defining the processing window required to obtain fully connected conductive tracks. (e) Representative line patterns together with cross-sectional views, revealing the geometry and vertical profile of the printed traces. (f) Current–voltage characteristics comparing a line printed from 15 metal droplets with a bulk metal conductor of identical length and diameter. (g) Photograph of a fully printed 555 timer circuit on a polycarbonate substrate. (h) Power-on demonstration of the printed circuit, verifying its functional integrity. (i) Residual footprints left on the polycarbonate surface after trace removal. (j) Bottom-side

morphology of a metal trace detached from the substrate. (k, l) Schematics and images showing the recovery and reuse of printed metal traces for recycling. (m) Conductive lines with three-dimensional crossovers fabricated using recycled metal material. (n) Enlarged view of the printed crossover structures. (o, p) Power-on tests of the three-dimensional crossover circuit. Reproduced from ref. [200]; Copyrights Elsevier (2021).

A variety of line patterns and geometries were successfully produced, with cross-sectional images revealing partial embedding of the metal into the softened polymer (Fig. 13e). This interfacial configuration provides strong mechanical adhesion while preserving the ability to detach the metal during recycling. Electrical characterization confirms that the printed lines exhibit ohmic I–V behavior comparable to bulk metal conductors of similar dimensions (Fig. 13f). This indicates minimal interfacial resistance between adjacent droplets and efficient electron transport across the printed traces. The practical capability of the method is demonstrated by printing a fully functional 555 timer circuit on a polycarbonate substrate (Fig. 13g), which operates reliably under power-on testing (Fig. 13h). This confirms that the printing approach is suitable for real electronic systems rather than simple demonstrators. Interfacial features after trace removal were also examined. The footprints left on the plastic substrate (Fig. 13i) and the bottom morphology of the detached metal traces (Fig. 13j) provide direct evidence of intimate metal–polymer contact during printing, while also explaining the clean separation during recycling. Crucially, the removed metal was remelted and reused to print new conductive lines (Fig. 13k–l), demonstrating true material recyclability without performance degradation. Using recycled metal, three-dimensional crossover structures were fabricated (Fig. 13m–n), and successful power-on tests confirmed the reliability of these complex circuits (Fig. 13o–p). Overall, this work shows that partial embedding of molten metal droplets enables a rare combination of mechanical robustness, high electrical performance, and recyclability. By engineering the molten metal–polymer interface to be strong yet reversible, the study establishes a compelling pathway toward sustainable, reconfigurable, and environmentally responsible printed electronics.

6.3. Critical insights, limitations, and future directions

- Confinement fundamentally transforms molten metal behavior by shifting dominance from bulk fluid dynamics to surface tension, capillarity, and interfacial mechanics, enabling stable droplets, wires, and films at micro- and nanoscale dimensions.
- The presence of an ultrathin oxide skin allows molten metals to maintain non-equilibrium geometries and provides mechanical resilience, enabling stretchable, reconfigurable, and self-healing interfaces in confined environments.
- Micro- and nanoscale confinement suppresses turbulence and limits mass transport, thereby stabilizing otherwise highly reactive molten metals and enabling their integration into microelectronics, microfluidic systems, and nanoscale devices.
- Laser-induced interfacial stripping mechanisms reveal that molten metal removal can be stress-driven rather than evaporation-driven, providing higher precision and lower energy consumption in advanced manufacturing processes.
- In soft matter systems, the mutual adaptability between molten metals and polymers enables mechanically interlocked interfaces where the polymer deforms while the liquid metal maintains electrical continuity.
- Partial embedding of molten metal droplets into thermoplastic substrates demonstrates a unique combination of electrical performance, mechanical stability, and recyclability, opening new directions for sustainable electronics.
- Confined molten metal systems are highly sensitive to interfacial defects, thermal gradients, and stress localization, which can lead to instability or

structural failure.

- The stability of molten metal interfaces strongly depends on precise control of oxidation and oxide-skin formation, yet these processes are difficult to regulate under dynamic thermal and environmental conditions.
- Many demonstrated systems operate within narrow processing windows, where slight deviations in temperature, droplet dynamics, or substrate properties can significantly affect interfacial behavior.
- Long-term reliability remains uncertain because repeated deformation, thermal cycling, and oxidation can degrade metal–polymer interfaces over time.
- Current fabrication strategies often rely on laboratory-scale printing or microfabrication techniques, limiting scalability for industrial manufacturing.
- The coupling between molten metals and soft substrates introduces complex mechanical interactions, which are not yet fully understood at the molecular or mesoscale levels.
- Development of advanced in situ imaging and spectroscopic techniques is needed to directly observe molten metal behavior under confined and soft-matter conditions.
- Future studies should focus on engineering oxide skins and interfacial chemistries to achieve controllable adhesion, stability, and electronic performance.
- Integration of molten metals with 2D materials, porous frameworks, and functional polymers may enable new hybrid systems with improved

mechanical and electrical properties.

- Designing self-regulating interfaces that dynamically adjust oxidation state and interfacial stress could enhance reliability in flexible and wearable devices.
- Scalable manufacturing approaches, including high-precision droplet printing and microfluidic templating, will be essential for translating laboratory demonstrations into practical technologies.
- Long-term studies on mechanical fatigue, thermal stability, and environmental durability are necessary to ensure the reliability of molten metal interfaces in real-world applications.

7. Emerging applications driven by interfacial science

The interfacial science of molten metals has emerged as a crucial driver of innovation in modern materials engineering, reshaping how we design, process, and apply metals in real-world technologies [201]. Unlike bulk properties that have historically dominated metallurgical research, interfaces (where liquid metal meets gas, solid, or another liquid) are the true origin of many critical behaviors such as wettability, surface tension, impurity segregation, and nucleation during solidification. These interfacial phenomena critically influence casting, additive manufacturing, alloy formation, and metal refining, with inadequate interfacial control leading to defects such as porosity, cracks, weak adhesion, and unwanted inclusions. Consequently, interfacial science is not merely an academic pursuit but a fundamental requirement for improving the reliability and performance of metal-based systems [202]. Emerging applications enabled by interfacial control span across high-impact sectors. In advanced casting and solidification, controlled interfacial reactions allow grain refinement and defect-free components that are essential for aerospace and automotive industries. In metal additive manufacturing, interface engineering ensures melt pool stability, improved wetting of powders, and reduced

porosity, leading to stronger and lighter parts. In energy storage, particularly liquid metal batteries, interfaces determine ion transport, stability of liquid layers, and prevention of dendrite formation, which are vital for achieving high capacity and long cycle life. Furthermore, interfaces play a decisive role in corrosion resistance and the development of metal-ceramic composites, where controlled interfacial bonding can produce hybrid materials with exceptional strength and thermal stability. Beyond technical benefits, the importance of interfacial science is deeply human: every premature material failure in infrastructure, transportation, or energy systems carries significant costs in terms of safety, resources, and economic loss [203].

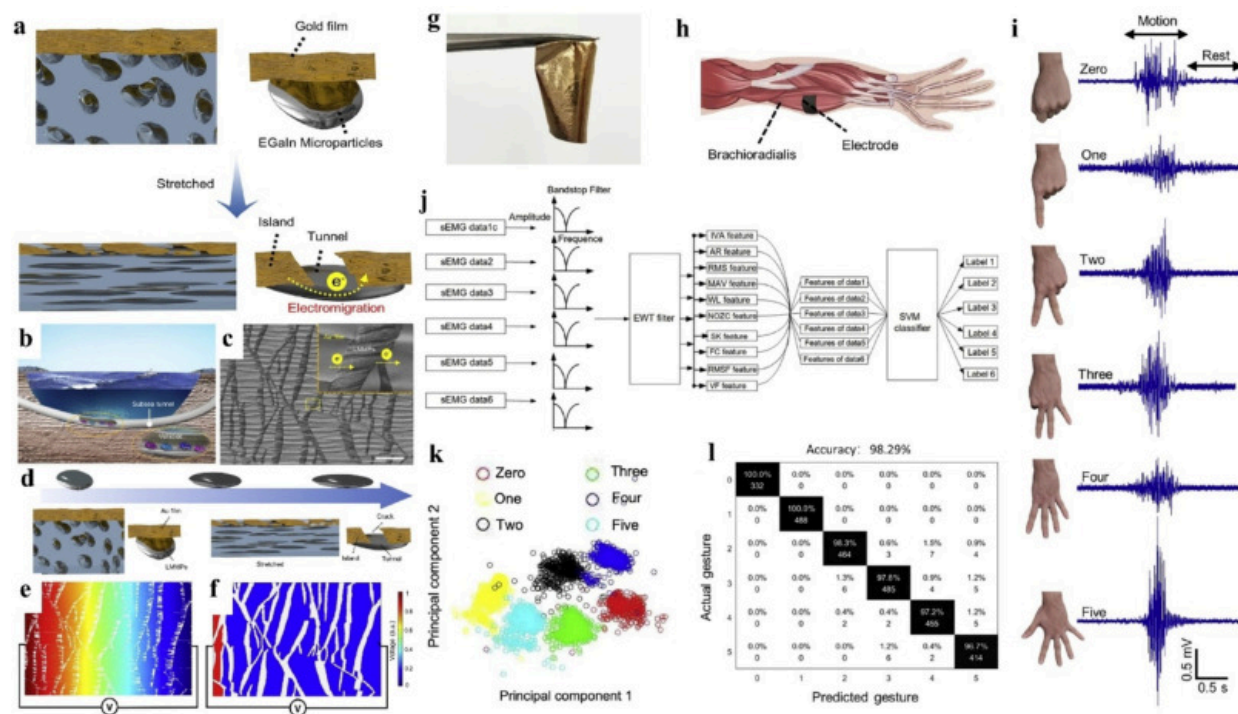
7.1. Soft electronics and stretchable conductors

Recent advances focus on interfacial engineering strategies that stabilize liquid metal particles within soft substrates and promote controlled connectivity. One common approach is to embed liquid metal microparticles into elastomeric polymers such as Polydimethylsiloxane (PDMS) and then apply a thin metallic film to form hybrid architectures. This design creates a dual conductive system where the surface film provides low-resistance pathways under small strain, while the liquid metal particles act as adaptive “bridges” that maintain conduction when the film cracks under large deformation. Such architectures exploit the mismatch in mechanical properties between rigid metallic films and fluid particles, enabling reliable electrical performance up to high strain levels. Additionally, interfacial oxide layers formed on liquid metals play a crucial role in maintaining particle stability and adhesion within polymers, thereby enhancing durability and mechanical robustness [204]. Applications of these interfacially-engineered liquid metal conductors include stretchable interconnects, soft sensors, wearable electrodes, and human-machine interfaces. For example, liquid metal-embedded electrodes have been successfully used for surface electromyography (sEMG) and gesture recognition, demonstrating high sensitivity and stable signal transmission under repeated deformation. Overall, the unique combination of fluidic metal behavior and interface-driven connectivity makes molten metals a promising platform for next-

generation soft electronics, where flexibility, reliability, and electrical performance must coexist.

To validate this idea, Li et al. reported a stretchable conductor design based on liquid metal microparticles (LMMPs) embedded in PDMS and covered by a thin Au film, creating a conductive network that remains functional under large deformation [205]. The authors proposed an “island–tunnel” architecture where the Au layer forms rigid islands while the LMMPs provide continuous conductive tunnels, enabling stable electron transport despite cracking of the Au film (Fig. 14a). The working mechanism was explained using an analogy of a subsea tunnel connecting two landmasses, where the Au islands represent the mainland and the LMMPs represent the tunnel, allowing continuous transport even when the surface layer is broken (Fig. 14b). The microstructure of the conductor under 40% strain shows that the Au film develops cracks while LMMPs bridge the gaps, maintaining electrical connectivity in a manner analogous to the subsea tunnel model (Fig. 14c). The morphology changes before and after stretching further confirms that the Au film breaks into islands while LMMPs deform and form conductive bridges across cracks (Fig. 14d). Voltage distribution simulations based on the SEM image demonstrate continuous voltage distribution when LMMPs are present, indicating stable electrical conduction under strain (Fig. 14e). In contrast, eliminating LMMPs results in a rapid voltage drop, confirming that the Au islands alone cannot maintain electrical connectivity and that LMMPs are essential for conduction (Fig. 14f). A flexible sEMG electrode based on the LMMP–Au conductor was fabricated to demonstrate practical application (Fig. 14g). The electrode was applied to record sEMG signals from the brachioradialis muscle, demonstrating reliable muscle activity detection (Fig. 14h). The recorded sEMG waveforms show distinct patterns corresponding to different hand gestures, indicating clear signal differentiation (Fig. 14i). The machine learning workflow used for gesture classification is illustrated, showing data processing, feature extraction, and classification steps (Fig. 14j). Principal component analysis of the recorded sEMG signals reveals well-separated clusters for each gesture, confirming the discriminative nature of the signals (Fig. 14k). The confusion matrix of the predicted results shows high classification

accuracy, with white values indicating correct predictions and black values indicating incorrect classifications (Fig. 14l). They verified that interfacial engineering of LMMP–Au structures can enable highly stretchable conductors with stable electrical performance by providing a promising strategy for soft electronics and wearable human–machine interfaces.



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Fig. 14. (a) Schematic of the fabrication strategy: a soft PDMS substrate is embedded with LMMPs and coated with a thin gold film, creating a highly stretchable conductor. (b) A conceptual diagram illustrating a subsea tunnel connecting two mainland regions. (c) SEM image of the conductor under 40% tensile strain. The cracked Au film represents the mainland, while the LMMPs act as the subsea tunnel, allowing electrons (analogous to vehicles) to pass through despite surface fractures. Scale bar: 100 μm . (d) Schematic illustration showing the morphology of the LMMPs and Au film before and after

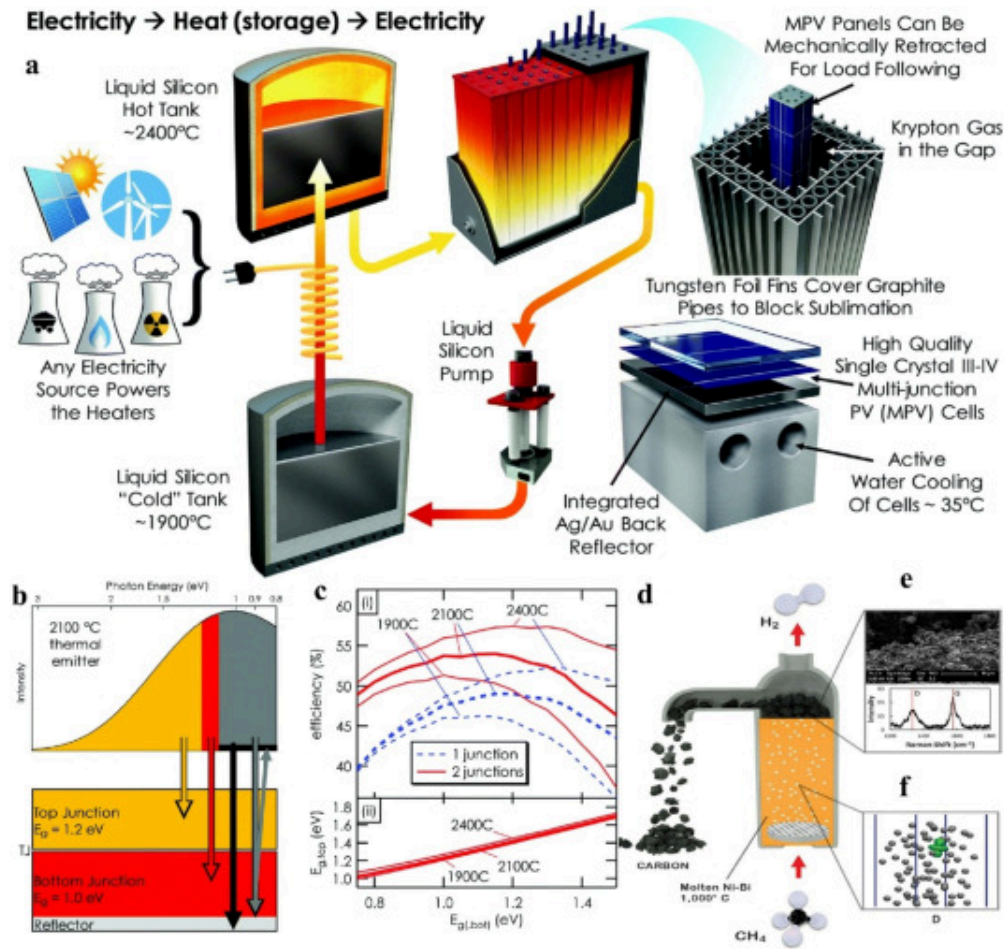
stretching. (e–f) Voltage distribution simulations based on the SEM image in (c) with (e) LMMPs retained and (f) LMMPs removed. (g) Photograph of the flexible sEMG electrode based on the LMMP–Au conductor. (h) The electrode attached to the brachioradialis muscle to record sEMG signals. (i) Representative sEMG waveforms corresponding to different hand gestures. (j) Illustration of the machine learning workflow used for hand gesture classification. (k) Principal component analysis of the recorded sEMG signals, showing clear separation of gesture clusters. (l) Confusion matrix of the predicted hand gestures, where white and black values represent the percentages of correct and incorrect classifications, respectively. Reproduced from ref. [205]; Copyrights Elsevier (2022).

7.2. Energy conversion and storage

Molten metals provide a compelling platform for energy conversion and storage by enabling electricity to be stored in the form of high-temperature sensible heat rather than through chemical reactions [206]. This strategy mitigates degradation and extends operational lifetime compared to conventional electrochemical batteries, rendering molten-metal systems highly suitable for large-scale, long-duration energy storage. Owing to their high thermal conductivity and heat capacity, molten metals can absorb and release energy rapidly while maintaining uniform temperature profiles, which is essential for efficient charging and discharging processes [207]. Their ability to operate at extreme temperatures also allows effective coupling with advanced thermal-to-electric conversion technologies, such as thermophotovoltaic systems, where stored heat can be reconverted into electricity with high theoretical efficiency [208]. In addition, the use of low-cost, earth-abundant molten materials improves economic feasibility and scalability [209]. However, the performance and durability of these systems are strongly governed by interfacial interactions between molten metals, containment materials, and energy conversion components, highlighting the need for careful materials selection and interface engineering.

Amy et al. demonstrated an innovative molten-material–based energy storage and reconversion concept that leverages extreme-temperature sensible heat storage coupled

with photonic energy recovery [210]. In this approach, electrical energy from any source is first converted into thermal energy through Joule heating and stored in a liquid medium operating at ultrahigh temperatures (1900–2400 °C) (Fig. 15a). A key strength of the system lies in the use of low-cost molten materials, such as metallurgical-grade silicon, which significantly reduces material costs while enabling large-scale thermal storage with remarkably low heat loss ($\sim 1\%$ per day). Unlike conventional electrochemical storage, this strategy bypasses degradation-prone interfaces by relying on bulk thermal storage, making it inherently scalable and long-lived. When electricity is needed, the stored thermal energy is released by pumping the hot liquid through an array of radiative tubes that emit intense thermal radiation, effectively converting stored heat into optical energy.



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Fig. 15. (a) Schematic of the proposed molten-material-based energy system. (b) Illustration of photon absorption from a 2100 °C thermal emitter in a two-junction PV cell, assuming 98% reflectivity for sub-bandgap photons (gray) and 2% absorption in the back reflector (black); TJ denotes the tunnel-junction interconnect. (c) Modeled efficiencies of single- and two-junction PV cells at emitter temperatures of 1900–2400 °C as a function of the bottom-junction bandgap, with the optimal top-junction bandgap selected for maximum efficiency (inset). Reproduced from ref. [210]; Copyrights Royal

Society of Chemistry (2019). (d) Schematic of a molten-metal bubble-column reactor for methane conversion into hydrogen and solid carbon with continuous carbon removal. (e) SEM image and Raman spectrum of the produced carbon, showing characteristic D (1350 cm^{-1}) and G (1582 cm^{-1}) bands. (f) Ab initio molecular dynamics simulation illustrating the electronic orbital (green) of a Pt atom dissolved in a molten Bi (gray) alloy. Reproduced from ref. [211]; Copyrights Science (2017).

The energy reconversion mechanism is further detailed through the photovoltaic coupling scheme (Fig. 15b). They employed multi-junction photovoltaic (PV) cells designed to selectively absorb high-energy photons in the visible and near-infrared range emitted by a $\sim 2100\text{ }^{\circ}\text{C}$ thermal source. Photons with energies below the bandgap are largely reflected (assumed reflectivity $\sim 98\%$) back to the emitter using a highly reflective rear mirror, thereby minimizing parasitic absorption and improving overall system efficiency. This photon recycling strategy is critical, as it reduces thermal losses and enhances the thermophotovoltaic conversion efficiency. The inclusion of tunnel junctions ensures efficient series interconnection between subcells, allowing optimal charge transport across the multi-junction architecture. The performance limits and design trade-offs of this system are quantified through detailed efficiency modeling (Fig. 15c). They systematically evaluated one- and two-junction PV cells across emitter temperatures ranging from 1900 to $2400\text{ }^{\circ}\text{C}$, correlating conversion efficiency with the bottom-junction bandgap. For two-junction cells, the top-junction bandgap was optimized for each bottom-junction configuration to maximize efficiency. These results demonstrate that carefully matched PV architectures can achieve substantially higher efficiencies under extreme thermal emission conditions compared to single-junction devices.

From a critical perception, their study presents a compelling shift away from conventional battery-based storage toward molten-material thermal systems that exploit fundamental radiative and interfacial physics. The concept elegantly integrates molten materials, high-temperature heat transfer, and photonic energy conversion, addressing scalability and cost limitations inherent to electrochemical storage. However, practical implementation

will require rigorous control over material stability, containment at extreme temperatures, and long-term durability of PV cells under intense thermal radiation. Despite these challenges, they provide a strong conceptual and quantitative foundation for next-generation molten-material energy systems, illustrating how interfacial control between molten media, radiative surfaces, and photovoltaic absorbers can enable efficient, large-scale energy storage and recovery.

Similarly, Upham et al., in a landmark study published in *Science*, proposed a molten-metal-based reactor concept for methane conversion that enables hydrogen production without direct CO₂ emissions [211]. In this system, methane is decomposed in a molten-metal bubble-column reactor, where the molten phase serves simultaneously as a heat-transfer medium, reaction environment, and carbon sink (Fig. 15d). Methane gas is injected into the molten metal, forming bubbles that rise through the liquid and thermally decompose into hydrogen gas and solid carbon. A main benefit of this configuration is the continuous removal of carbon from the reaction zone, which prevents catalyst deactivation and reactor clogging (key challenges in conventional methane pyrolysis systems). The molten metal plays a central role in stabilizing the reaction environment by providing uniform high-temperature conditions and suppressing unwanted side reactions. Unlike solid catalysts, which rapidly deactivate due to carbon fouling, the liquid metal interface allows carbon to nucleate, grow, and separate from the reactive zone without blocking active sites. This design fundamentally alters the gas–solid interaction problem by replacing it with a gas–liquid interface, where mass transfer and carbon management are inherently more controllable. From a progression perspective, this approach offers a scalable route to hydrogen production while generating solid carbon as a potentially valuable by-product rather than a waste stream. The nature of the produced carbon is revealed through microstructural and spectroscopic characterization (Fig. 15e–f). SEM shows that the carbon formed during methane decomposition exhibits a particulate morphology, indicating controlled nucleation and growth rather than uncontrolled deposition. Raman spectroscopy further confirms the structural characteristics of the carbon, with prominent D and G bands at approximately 1350 and

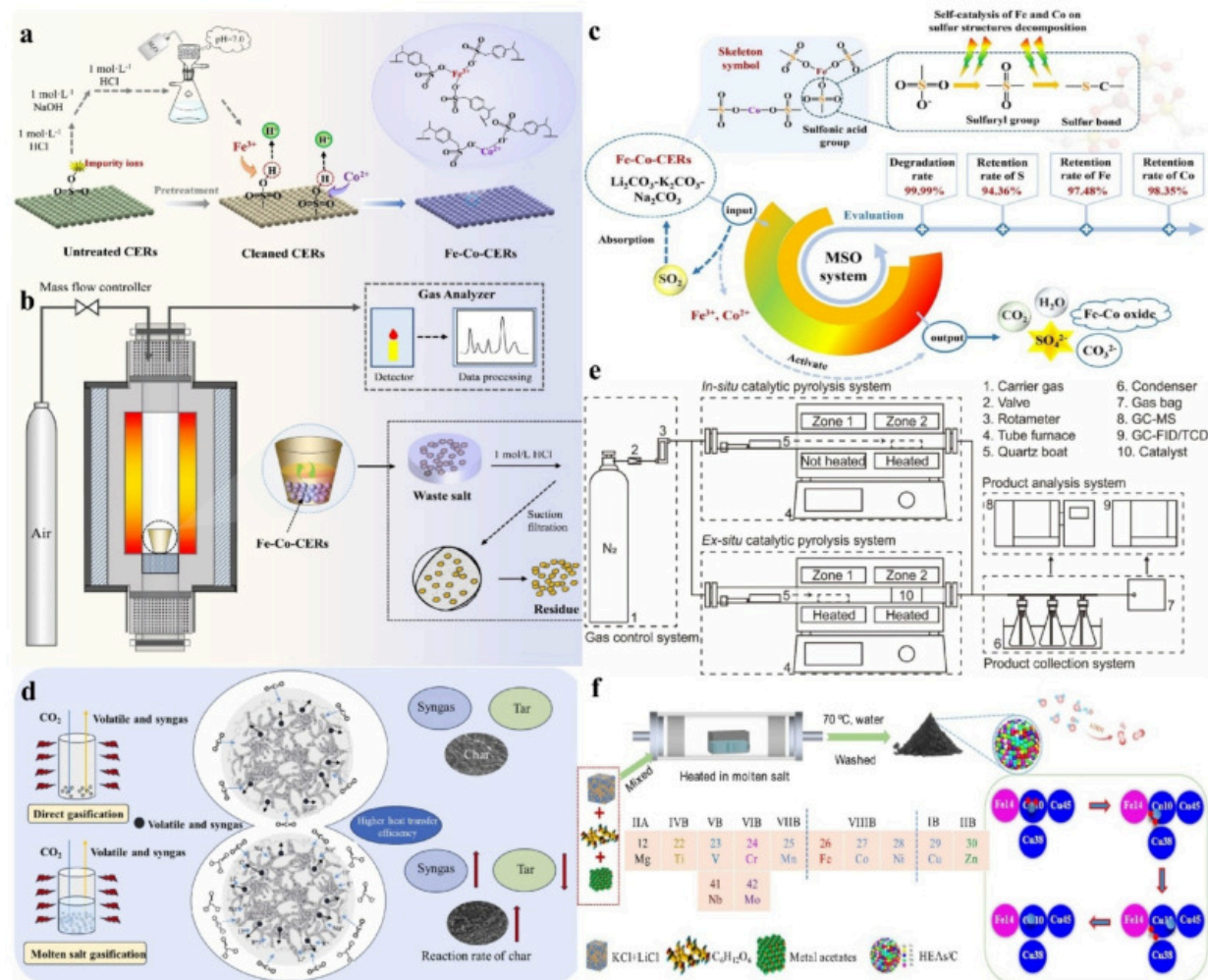
1582 cm^{-1} , respectively. The presence of these bands suggests predominantly disordered graphitic carbon, which could be suitable for downstream applications such as carbon black or functional fillers, depending on processing conditions. From a critical standpoint, their study presents a paradigm shift in methane conversion technology by leveraging molten-metal interfaces to decouple hydrogen production from carbon management. The elimination of CO_2 formation at the reaction stage directly addresses one of the most pressing challenges in hydrogen production. However, the long-term stability of the molten metal, control over carbon morphology, and energy requirements for maintaining high operating temperatures remain important considerations for industrial deployment.

7.3. Catalysis and chemical synthesis

Unlike conventional solid catalysts, molten metal environments provide a dynamic interface where reactants can continuously interact with dissolved or dispersed active species, enabling rapid reaction kinetics and improved selectivity [212]. Moreover, molten media can dissolve, redistribute, and regenerate catalytic components in situ, which mitigates catalyst deactivation and enables continuous operation. These distinctive properties render molten-metal systems highly suitable for demanding chemical processes, including biomass conversion, gasification, and selective oxidation, which require elevated temperatures and aggressive reaction environments. The molten-metal-assisted catalysis opens new possibilities for scalable chemical manufacturing by leveraging interfacial control and fluidic reaction pathways.

In recent 2025, Liu et al. conducted a systematic study focused on converting untreated cation-exchange resins (CERs) into highly active Fe–Co–modified CERs through a stepwise pretreatment and metal coordination strategy [213]. The process begins with impurity removal and structural cleaning of the raw CERs to expose active sites and enhance surface accessibility (Fig. 16a). Subsequently, Fe^{3+} and Co^{2+} ions are introduced via ion exchange, which allows them to anchor firmly onto the CER framework through stable coordination bonds. This step is crucial because it prevents metal leaching at high temperatures and ensures long-term catalytic stability. The resulting Fe–Co–modified

CERs show improved catalytic activation due to the synergistic interaction between Fe and Co species, which enhances redox cycling and oxygen activation during the reaction. The experimental setup for molten-salt-assisted catalytic processing ([Fig. 16b](#)). They used a high-temperature reactor with controlled gas flow to create a stable molten-salt environment, enabling efficient contact between reactants and active catalytic sites. Product separation and post-reaction recovery are incorporated into the design to ensure that the Fe–Co–CERs can be retrieved and reused, demonstrating the practicality of the approach. In this system, molten salt acts as a heat transfer medium, a solvent for ionic species, and a protective matrix that maintains catalyst integrity under severe operating conditions. The molten salt oxidation system efficiently absorbs SO_2 and uses the Fe and Co species to self-catalyze the decomposition of sulfur-containing structures ([Fig. 16c](#)). This results in high degradation efficiency while selectively retaining S, Fe, and Co in the system. The MSO process highlights how molten metal/salt interfaces can enable simultaneous pollutant removal and catalyst regeneration, offering a sustainable pathway for chemical conversion without generating secondary waste.



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Fig. 16. Catalytic functions and chemical synthesis pathways enabled by molten salt and molten-metal-assisted systems. (a) Stepwise pretreatment and metal coordination strategy for converting untreated CERs into Fe–Co-modified CERs, illustrating impurity removal, ion exchange, and stable Fe³⁺/Co²⁺ anchoring for catalytic activation. (b) Experimental setup for molten-salt-assisted catalytic processing, including controlled gas flow, high-temperature reactor, product separation, and post-reaction recovery of Fe–Co-

CERs and residues. (c) Schematic representation of the MSO system, highlighting SO₂ absorption, self-catalyzed decomposition of sulfur-containing structures by Fe and Co species, and the high degradation efficiency with selective retention of S, Fe, and Co. Reproduced from ref. [213]; Copyrights Elsevier (2025). (d) Comparison between direct gasification and molten-salt gasification routes. Reproduced from ref. [214]; Copyrights Elsevier (2025). (e) In-situ and ex-situ catalytic pyrolysis systems integrated with molten salt media. Reproduced from ref. [215]; Copyrights Elsevier (2025). (f) Molten-salt-assisted synthesis and transformation of multicomponent metal. Reproduced from ref. [216]; Copyrights Elsevier (2025).

In addition, She et al. compared conventional direct gasification with molten-salt-mediated gasification [214], demonstrating that molten salts significantly improve heat transfer and mass transfer during the reaction (Fig. 16d). In the molten-salt route, char conversion is accelerated because the molten medium enhances contact between carbonaceous solids and reactive gases, leading to higher syngas yield and reduced tar formation. The molten salt effectively dissolves and transports intermediate species, preventing tar condensation and blocking. This results in cleaner gas products and more efficient conversion, making the molten-salt route a powerful strategy for upgrading biomass or waste feedstocks into syngas with improved energy efficiency. Furthermore, Wang et al. developed integrated catalytic pyrolysis systems where molten salt plays an active role in controlling reaction conditions and product formation [215]. Their setup includes in-situ and ex-situ catalytic configurations, with separate heating zones and dedicated pathways for gas and liquid product collection (Fig. 16e). The molten salt medium enables uniform heating and stabilizes reactive intermediates, which improves selectivity and yields of desired products. The design also allows precise evaluation of reaction products through continuous gas analysis and controlled sampling, demonstrating how molten media can enhance both reaction performance and analytical accuracy. Moreover, Zhang Wang et al. reported a molten-salt-assisted synthesis route for multicomponent metal species, where thermal treatment and controlled washing enable redistribution and aggregation of transition metals into catalytically active clusters and

alloys [216]. During thermal processing in molten salt, metal ions migrate and form stable clusters due to high mobility and reduced diffusion barriers (Fig. 16f). Subsequent washing removes excess salts while preserving the newly formed metallic structures. This method enables the controlled generation and regeneration of active catalytic phases, making it a influential approach for designing high-performance catalysts with tunable composition and structure.

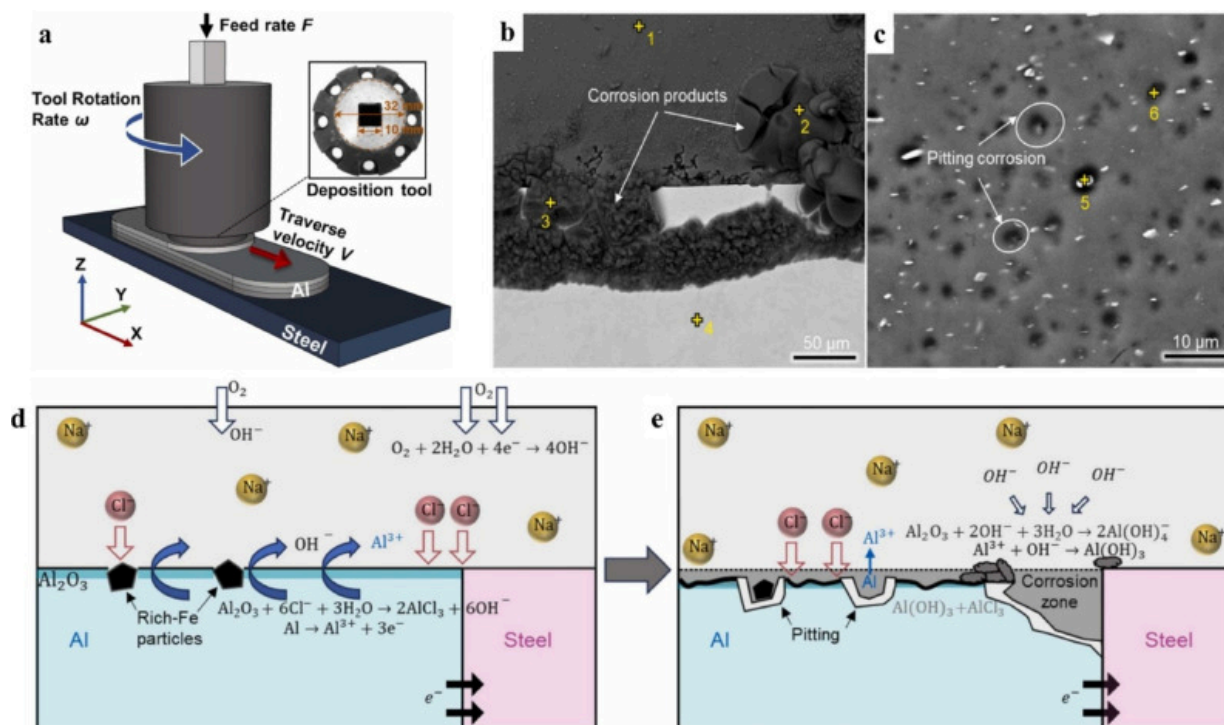
7.4. Additive manufacturing and joining technologies

Additive manufacturing (AM) and joining technologies involving molten metals are revolutionizing how complex structures and multi-material components are fabricated [217], [218]. Unlike traditional casting or welding, AM enables layer-by-layer material deposition, allowing precise control over geometry, microstructure, and interface formation [219]. Laser additive manufacturing has transformed industrial production, but conventional Gaussian lasers often cause high thermal gradients, porosity, and coarse grains. Advanced strategies such as beam shaping, laser oscillation, and multi-field assistance improve melt pool stability, refine grains, and suppress defects [220]. When joining dissimilar metals, the key challenge lies in controlling interfacial reactions, metallurgical bonding, and residual stress, specifically under high-temperature conditions [221]. In this context, friction-based solid-state deposition techniques, such as additive friction stir deposition (AFSD), are increasingly attractive because they avoid full melting, reduce thermal distortion, and promote strong metallurgical bonding through plastic deformation and dynamic recrystallization. These methods are particularly valuable for creating hybrid joints like Al/steel, where dissimilar thermal expansion and chemical compatibility often cause severe interfacial defects. In 2024, numerical and experimental investigations have demonstrated that careful control of laser welding parameters significantly affects molten pool stability and post-weld deformation in aluminum alloy joints. For example, simulations of 6056 Al T-joints using finite element software revealed that bilateral synchronous welding enhances molten pool stability compared to asynchronous welding, and optimized welding sequences can minimize post-weld stress and deformation [222]. Additionally, Understanding the atomization

mechanisms and defect formation in alloy powders is critical for high-quality additive manufacturing. Dual-gas nozzle atomization, coupled with droplet breakup models, has been shown to control particle size distribution, minimize satellite formation, and improve powder morphology, ultimately enhancing part performance in AM processes [223]. Complementary studies on aluminum alloy additive manufacturing demonstrated that cold metal transfer (CMT) with arc oscillation broadens the molten pool, enhances fluidity, reduces porosity, and promotes secondary nucleation, leading to more homogeneous microstructures and improved mechanical properties such as tensile strength and elongation [224]. Collectively, these insights highlight the importance of controlling both powder atomization mechanisms and deposition/melting dynamics to achieve high-performance, defect-minimized metal additive manufacturing.

Moreover, Li et al. reported extruded Al alloy and annealed steel as base materials, and the initial microstructures [225]. The deposition tool was a hollow 40CrNi structural steel component with an inner square bore of $10 \times 10 \text{ mm}^2$ and a shoulder diameter of 32 mm (Fig. 17a). Before deposition, the substrate surfaces were sanded and cleaned with alcohol to remove contaminants, ensuring proper bonding during AFSD. The deposition process followed a reciprocating traveling strategy to form a single deposited layer with an average thickness of $\sim 1.75 \text{ mm}$. The optimized parameters included a rotation speed of 300 rpm, traversing velocity of 150 mm/min, and feed rate of 100 mm/min, which collectively ensured sufficient material plasticization and bonding without excessive heat input. The AFSD process setup, showing the deposition tool inserted into the build region and the coordinate system (Z: build direction, X: transverse direction, Y: normal direction). Moreover, SEM–BSE images of corrosion products formed at the Al/steel interface and on the Al side after 96 hours of immersion in 3.5 wt% NaCl solution. The corrosion products were primarily concentrated at the interface, indicating that interfacial microstructures and phases play a dominant role in corrosion initiation. EDS analysis revealed that the corrosion layer mainly contained Al, O, H, and Cl, suggesting the formation of aluminum oxides and chlorides such as Al_2O_3 , $\text{Al}(\text{OH})_3$, and AlCl_3 (Fig. 17b–c). Additionally, localized corrosion pits were observed in the Al matrix (Fig. 17c),

particularly around bright second phases identified as Al–Fe–Si intermetallics (composition ~83.4 wt% Al, 9.5 wt% Fe, 3.5 wt% Si, 3.6 wt% O). These phases acted as cathodic sites, creating micro-galvanic cells that accelerated localized dissolution of the surrounding Al matrix. The corrosion mechanism of AFSD Al/steel joints in saline solution, divided into two characteristic stages. In the selective dissolution initiation stage (Fig. 17d), the Al matrix begins to dissolve preferentially near cathodic intermetallic particles due to micro-galvanic corrosion, while corrosion products accumulate at the interface. As corrosion improves, the process enters the accelerated corrosion stage (Fig. 17e), where continuous anodic dissolution and enhanced galvanic coupling promote rapid corrosion propagation and pit growth. This stage is characterized by extensive interface degradation and localized attack around Fe-rich phases, which significantly compromises the integrity of the joint.



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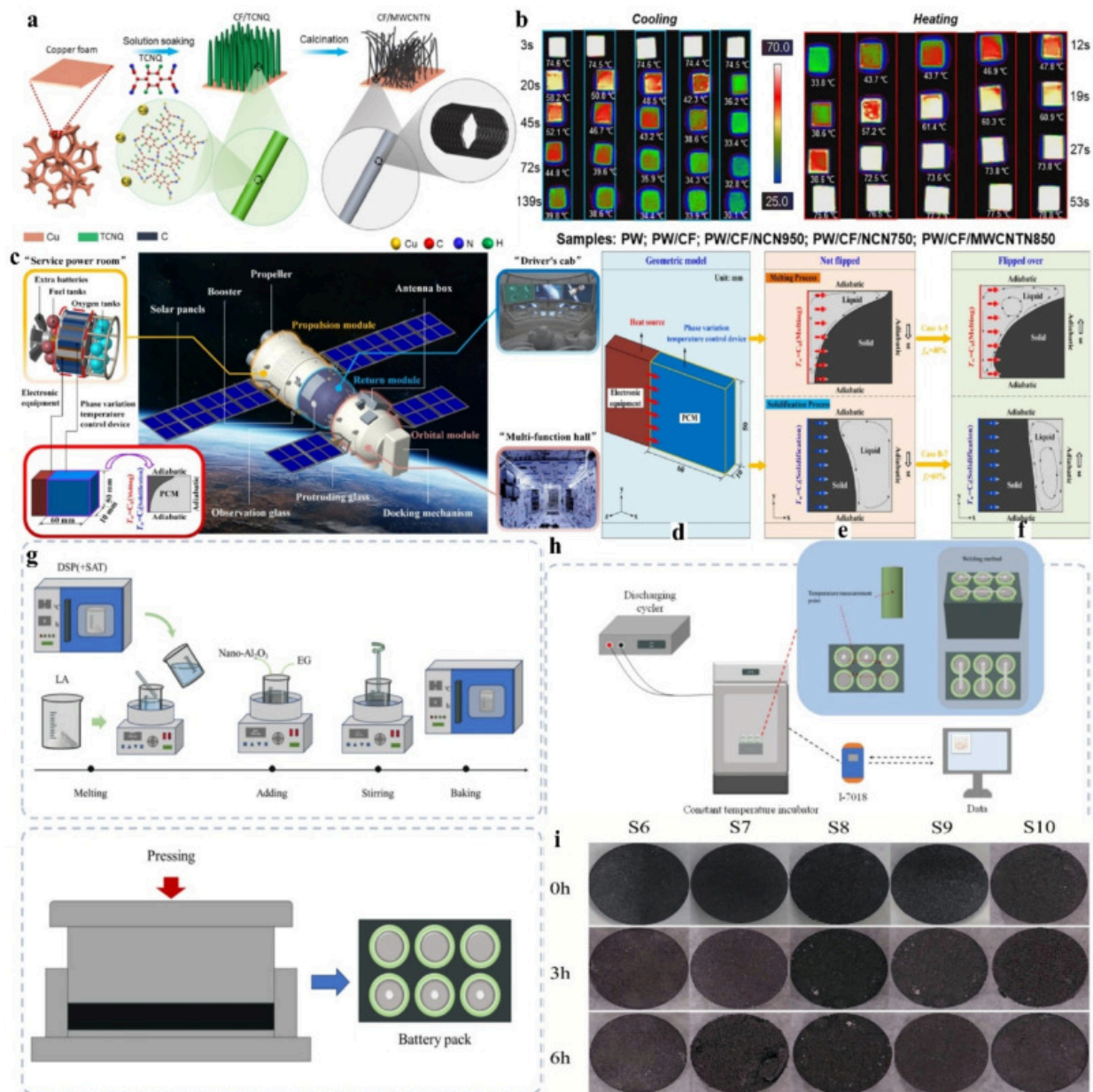
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Fig. 17. (a) Schematic illustration of the AFSD process, showing the tool end inserted into the build with the coordinate directions: Z (building direction), X (transverse direction), and Y (normal direction). (b–c) SEM–BSE images of corrosion products formed at the Al/steel interface and on the Al side after 96 h of immersion in 3.5 wt% NaCl solution. Schematic diagrams depicting the corrosion mechanism of AFSD Al/steel in saline solution, where (d) Represents the initial stage of selective dissolution and (e) shows the subsequent accelerated corrosion stage driven by coupled anodic and cathodic reactions. Reproduced from ref. [225]; Copyrights Elsevier (2025).

7.5. Thermal management and thermal storage

Efficient thermal management and thermal energy storage are central challenges in advanced energy systems, aerospace platforms, power electronics, and high-density batteries. Molten metals are particularly attractive in this context due to their exceptionally high thermal conductivity, low interfacial thermal resistance, and rapid heat redistribution enabled by metallic bonding in the liquid state. Unlike conventional organic phase-change materials, molten metals tolerate high heat fluxes, function at elevated temperatures, and provide rapid thermal buffering with minimal temperature gradients. Their interfacial behavior (governed by surface tension, oxide skins, and capillarity) further allows dynamic control of heat transfer, making them promising candidates for next-generation thermal regulation technologies. Although many current systems employ structured solid metals rather than fully molten phases, they often exploit the same fundamental principle: metals act as highly efficient heat transport networks [226], [227]. Recent (2024–2025) studies provide representative example of this concept. In 2025, Guo et al. reported copper is engineered into a three-dimensional porous foam that functions as a continuous metallic thermal backbone [228]. The foam is integrated with a carbon nanotube–derived network and subsequently infiltrated with a phase change medium, forming a thermally enhanced composite (Fig. 18a). The copper framework serves as a low-resistance pathway for heat flow, mimicking the rapid energy redistribution characteristic of molten metals. By establishing interconnected metallic conduction channels, the design suppresses internal thermal gradients and accelerates heat

propagation throughout the material. The thermal performance is directly visualized (Fig. 18b) through infrared thermography during heating and cooling cycles. Compared with pure paraffin, the composite containing the copper-based framework exhibits significantly faster temperature equilibration and more uniform thermal fields. The metallic network effectively reduces thermal lag, enhances melting and solidification kinetics, and stabilizes temperature fluctuations. This behavior parallels molten metal systems, where high electronic thermal conductivity enables rapid interfacial heat exchange and efficient thermal buffering under dynamic conditions. Thus, while the copper in this study remains structurally solid, its function as a thermally conductive metallic network reflects the same governing physics that underpin molten metal thermal management: rapid heat redistribution, minimized interfacial resistance, and enhanced phase transition control.



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Fig. 18. (a) Schematic illustration of the fabrication process for CF/MWCNTN composites. (b) Infrared thermal images comparing the heat transfer behavior of PW and PCCs. Reproduced from ref. [228]; Copyrights Elsevier (2025). (c) Overview of the spacecraft architecture, internal component configuration, corresponding 3D thermal control module model, and associated 2D boundary conditions. (d) Structural diagram of the phase-variation temperature control device with its 3D geometric dimensions. (e) Representation of the 2D computational domain and applied boundary conditions. (f) Structural state after flipping at a critical value. Reproduced from ref. [229]; Copyrights Elsevier (2024). (g) Schematic of material synthesis. (h) Experimental setup for battery performance testing. (i) Physical condition of the sample after testing. Reproduced from ref. [230]; Copyrights Elsevier (2025).

Li et al. also reported that advanced phase-variation thermal control devices can be strategically engineered to regulate heat flow in complex aerospace environments [229]. The overall spacecraft configuration and internal arrangement of thermal control components (Fig. 18c), where a three-dimensional model of the thermal regulation module is presented together with the corresponding two-dimensional boundary conditions used for simulation. The schematic highlights how heat-generating units are thermally coupled to a phase-based control system to ensure temperature uniformity under fluctuating thermal loads. The structural configuration and geometric parameters of the phase-variation temperature control device (Fig. 18d). The design integrates a thermally conductive framework with a phase change medium, allowing excess heat to be absorbed during peak operation and released during cooling. This configuration enables directional heat spreading and enhances system-level thermal buffering capacity. The computational domain and applied boundary conditions are further clarified (Fig. 18e), where heat flux inputs and environmental constraints are defined to model realistic working conditions. The simulations demonstrate how conductive pathways redistribute localized heat and suppress temperature gradients, thereby preventing hot-spot formation. The thermal response after structural flipping at a critical parameter value (f_m or f_s) (Fig. 18f). This reorientation modifies internal phase distribution and heat flow

pathways, revealing the adaptive nature of the device under changing thermal conditions. Such reconfigurable heat transfer behavior reflects principles similar to those observed in molten metal systems, where metallic conductivity and phase dynamics collectively govern efficient thermal regulation.

Moreover, Bai et al. reported in their 2025 study that metallic–phase hybrid systems can be strategically engineered to enhance thermal regulation in electrochemical energy storage devices [230]. The material synthesis route (Fig. 18g) where a thermally conductive metallic framework is integrated with functional components to construct a composite capable of efficient heat dissipation. The schematic emphasizes the formation of interconnected metallic pathways that facilitate rapid heat spreading, analogous to the high electronic thermal conductivity observed in molten metal systems. The experimental configuration for battery thermal testing (Fig. 18h). The setup monitors temperature evolution under realistic operating conditions, accounting for internal heat generation and the mitigation of thermal runaway risks. The metallic network serves as a heat redistribution medium, minimizing localized temperature accumulation and enhancing overall thermal stability. The post-test condition of the sample (Fig. 18i), demonstrating structural integrity and stable thermal performance after cycling. The preserved morphology indicates effective thermal buffering and mechanical robustness under repeated heating–cooling processes. This behavior illustrates the fundamental principle of molten metal thermal management: rapid heat conduction, efficient interfacial transfer, and dynamic thermal stabilization to mitigate failure under high heat flux conditions.

8. Current challenges and limitations

Despite the rapid expansion of molten metal research across fields such as catalysis, soft electronics, energy storage, and additive manufacturing, several fundamental and technological challenges continue to limit the broader deployment of these systems. Many molten metal platforms operate at elevated temperatures, which introduces significant engineering constraints related to thermal management, containment, and

material compatibility. Sustaining stable operation under such conditions requires robust reactor designs, corrosion-resistant structural materials, and efficient heat-transfer control, all of which increase system complexity and cost. Another persistent challenge lies in controlling interfacial phenomena. The surfaces of molten metals are highly dynamic and are readily modified by oxidation, adsorption of impurities, or alloying elements. While the formation of thin oxide skins can impart mechanical stability and enable structural manipulation of otherwise fluid metals, uncontrolled oxide growth may alter wetting behavior, hinder mass transport, and introduce variability in electrical or catalytic performance. Achieving precise control over oxide formation, interfacial energies, and wetting dynamics therefore remains a critical issue, particularly for applications that rely on stable liquid–solid or liquid–gas interfaces.

Material stability and compatibility also present significant limitations. Molten metals can interact aggressively with surrounding materials, leading to corrosion, dissolution, or interdiffusion at interfaces. These processes can compromise containment systems, degrade catalytic surfaces, and reduce long-term operational reliability. In catalytic environments, additional complications arise from impurity accumulation, carbon deposition, or changes in alloy composition during operation, which can lead to catalyst deactivation and reduced selectivity. Mechanical and structural stability is another challenge, especially in emerging applications involving stretchable electronics and soft robotic systems. Maintaining continuous conductive pathways during repeated deformation requires careful control of oxide layers, particle dispersion, and liquid-metal confinement within flexible matrices. Without appropriate design strategies, mechanical fatigue, leakage, or particle agglomeration can degrade electrical performance and limit device durability.

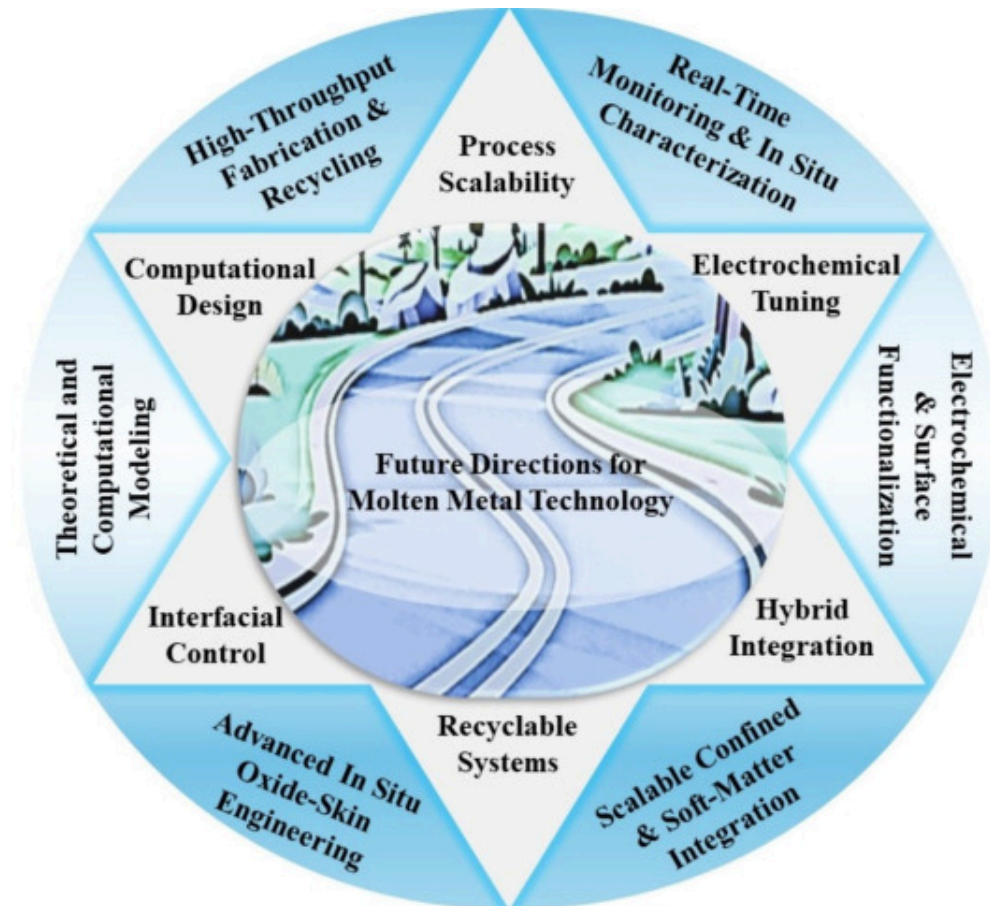
Scaling molten metal technologies from laboratory demonstrations to practical systems also remains nontrivial. Large-scale reactors, thermal storage systems, or manufacturing platforms must address issues related to heat transfer, flow control, and uniform temperature distribution. In addition, integrating molten metal components with other functional materials such as polymers, ceramics, photovoltaic devices, or electrochemical

cells requires reliable sealing, compatible thermal expansion behavior, and long-term structural integrity under operational stresses.

Finally, environmental, safety, and economic considerations cannot be overlooked. Some molten metal alloys involve elements that pose toxicity or handling concerns, and the management of high-temperature liquids introduces inherent safety risks. Furthermore, the cost of specialized alloys, containment materials, and energy-intensive processing may limit industrial adoption unless more sustainable and cost-effective alternatives are developed. Addressing these interconnected limitations will be essential for translating the unique physicochemical properties of molten metals into robust and scalable technologies across energy, manufacturing, and advanced functional materials.

9. Future perspective and research directions

Advancing molten metal technologies requires systematic strategies that address interfacial control, stability under extreme environments, and scalable system integration. Guided by the mechanistic insights discussed throughout this review, several priority research directions can be identified to accelerate progress in molten metal science and applications. In this review, we have summarized these directions ([Fig. 19](#)), highlighting key areas where targeted research and technological development are required to unlock the full potential of molten metal systems.



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Fig. 19. Key future directions for molten metal research and applications.

The following recommendations outline strategic research priorities for advancing molten metal science and technology:

- Develop advanced theoretical frameworks capable of describing the dynamic electronic structure, atomic diffusion, and interfacial energetics of molten metals. Multiscale simulations, including ab initio molecular

dynamics and machine-learning-assisted modeling, can help predict phase behavior, catalytic activity, and transport phenomena under realistic operating conditions.

- Establish controllable oxidation strategies to regulate the formation, stability, and mechanical properties of oxide skins on molten metal surfaces. Approaches such as controlled atmosphere processing, surface coatings, and dynamic oxide removal can enable stable wetting behavior and reliable electrical or catalytic performance.
- Investigate molten metal behavior under micro- and nanoscale confinement, particularly in polymer matrices, gels, and porous frameworks. Understanding how confinement affects wetting, capillary transport, and electrical connectivity will be essential for designing flexible circuits, sensors, and soft robotic components.
- Explore surface modification strategies through alloying, electrochemical tuning, and interfacial adsorption to tailor catalytic selectivity and electrochemical activity. Such approaches may enable molten metals to function as adaptable platforms for high-temperature electrochemistry, carbon conversion, and energy storage applications.
- Develop hybrid material systems that combine molten metals with ceramics, polymers, or molten salts to improve mechanical stability and interfacial adhesion. Rational composite design can mitigate leakage, enhance structural robustness, and expand the operational envelope of molten metal technologies.
- Improve reactor architectures, insulation systems, and containment materials capable of sustaining high-temperature molten environments while minimizing heat loss and material degradation.

- Develop manufacturing approaches such as liquid-phase printing, extrusion-based processing, and additive manufacturing to enable controlled deposition and patterning of molten metals for large-area devices and functional structures.
- Establish standardized testing protocols for evaluating molten metal systems across laboratories while exploring abundant, low-toxicity metal compositions and recyclable molten phases to support environmentally responsible and economically viable deployment.

Collectively, these directions emphasize the importance of integrating interfacial science, computational design, and scalable engineering approaches to transition molten metal technologies from laboratory-scale demonstrations to reliable platforms for energy, electronics, and advanced manufacturing.

10. Conclusion

Molten metals have evolved from classical metallurgical materials into a versatile platform spanning interfacial science, soft robotics, catalysis, energy conversion, and advanced manufacturing. Their unique combination of fluidity, metallic bonding, thermal and electrical conductivity, and dynamic surface chemistry originates from the intrinsic electronic structure and thermophysical behavior of metals in the liquid state. Molten-metal behavior is governed not only by bulk properties but also by interfacial phenomena, including wetting, capillary-driven flows, Marangoni convection, self-healing, shape reconfiguration, and electrocapillarity. Surface oxide layers, interfacial energies, and confinement effects collectively determine whether molten metals function as stable conductors, reactive catalysts, or mechanically adaptable media. Confinement and size-dependent effects further enhance their reconfigurability and functional versatility. These interfacial and thermophysical properties open pathways for designing molten-metal systems with tunable reactivity, mechanical adaptability, and energy-handling

capabilities. By integrating mechanistic insights from metallurgy, surface science, and materials engineering, this review establishes a unified framework for understanding molten-metal interfaces. Future advancements will require multiscale modeling, targeted oxide-skin engineering, soft-matter integration, electrochemical functionalization, and scalable fabrication and recycling strategies. Harnessing the interplay between liquid-state physics, interfacial energetics, and surface chemistry will be critical to translating molten metals from a fundamental research platform into reliable, scalable, and high-performance technologies across electronics, energy, and advanced manufacturing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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